

Automated BIM generation from inconsistent multi-view raster architectural drawings with missing dimensions

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ABSTRACT

Automated Building Information Model (BIM) generation from 2D architectural drawings faces critical challenges: single-drawing approaches generate incomplete models, undefined multi-view object correspondence hinders integration, and missing dimension annotations prevent reliable scale calibration. This paper proposes a two-phase method for automated BIM generation. The first phase performs drawing recognition and dimensional calibration using deep learning and a large multimodal model (LMM)-based reasoning framework to extract scale information. The second phase establishes object-level correspondence between floor plans and elevations using a cost-matrix and the Hungarian algorithm. Validated on 12,698 floor plans and 3400 elevations, the method achieved mAP@0.5 of 0.934 for floor plans and 0.838 for elevations, outperforming baselines by 41.8% and 16.2%. LMM-based scale recognition achieved 0.707 accuracy, surpassing traditional Optical Character Recognition (OCR). Cross-view matching achieved an F1-score of 0.93 and BIM generation achieved 0.693 volumetric Intersection over Union (vIoU), validating the multi-view drawing integration for information-consistent automated BIM generation.

1. Introduction

Building Information Models (BIMs) are rich information sources that digitally represent building assets and support a wide range of applications throughout the building lifecycle [1]. These applications include operation and maintenance (O&M), facility management [2], and energy performance simulation [3]. Despite these benefits, BIM data are rarely available for existing buildings, particularly for large-scale apartment complexes that account for a significant portion of residential stock in many developed countries. As many of these buildings approach the end of their service life, there is an urgent demand for renovation, retrofitting, and material reuse. However, architectural information for such buildings often exists only in paper-based or image-based 2-dimensional (2D) drawings [4,5], which lack the structured data required for digital analysis and management. The absence of BIMs not only interrupts energy-efficient retrofitting but also hinders large-scale carbon reduction initiatives.

Current practices in creating BIMs for existing buildings can be broadly categorized into two approaches. First, Scan-to-BIM approaches have been widely explored to create as-built or as-is BIMs that accurately

reflect the current state of buildings. While highly precise, these approaches require costly scanning equipment and expert personnel, posing economic and technical barriers to large-scale adoption [6]. In contrast, 2D drawing-based modeling represents the as-designed condition of buildings and plays an equally important role in 3-dimensional (3D) model generation. As-designed BIMs serve as essential references for renovation planning, regulatory compliance, while also providing a reference model for registering and aligning scan data in hybrid Scan-to-BIM workflows [7]. Yet, such models are rarely available for older apartment buildings, and their 3D generation often relies on manual digitization of drawings [8]. Although 2D drawing-based modeling is accessible and cost-effective, it remains labor-intensive, time-consuming, and prone to human error, underscoring the urgent need for automation [9].

In generating BIM models from 2D drawings, it is essential to recognize that each drawing type conveys complementary aspects of architectural information within the hierarchical data structure defined by the Industry Foundation Classes (IFC). In this hierarchy, entities such as *IfcBuilding*, *IfcBuildingStorey*, and *IfcBuiltElement* represent progressively detailed spatial and physical components of a building [10]. Floor

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plans primarily define horizontal geometry, including the length and width of elements and the placement of walls, doors, and windows, while elevation drawings specify vertical information such as object height, building elevation, and reference height. Certain relational entities, including *IfcRelVoidsElement* and *IfcRelFillsElement*, depend on both drawings to describe how openings are created and filled within host elements. Accordingly, floor plans and elevations collectively provide the geometric and semantic parameters necessary to instantiate complete *IfcBuiltElement* objects within the BIM hierarchy. Relying on a single drawing type results in incomplete or geometrically inconsistent 3D models, whereas accurate BIM generation requires the integrated interpretation of information represented in plans and elevations within the IFC-based hierarchical structure, as illustrated in Fig. 1.

To produce a complete 3D model from 2D drawings, a system must (1) accurately detect building objects within 2D drawings, (2) integrate complementary information from floor plans and elevations, and (3) map the detected components to corresponding BIM elements with appropriate spatial and dimensional attributes. These procedures are essential regardless of whether the process is manual or automated, as they define the logical sequence of generating 3D information from 2D representations. Recent studies have made significant progress toward this goal using artificial intelligence techniques. Machine learning and deep learning-based methods have been widely explored to detect architectural objects in floor plans and convert them into BIM elements. However, most of these approaches rely solely on floor plans, which provide horizontal spatial layouts but lack vertical attributes such as height. As a result, vertical dimensions are often predefined or manually input [11,12]. A few studies have investigated elevation-based approaches, but these approaches tend to depend on Computer-Aided Design (CAD) metadata such as layer structures and fail to capture the internal spatial layout of buildings, limiting their applicability to real-world legacy drawings [13]. These limitations highlight the importance of integrating complementary information from multiple drawing types to achieve accurate and scalable BIM generation, which motivates subsequent research efforts on multi-drawing integration.

Some studies have attempted to use multiple types of drawings, such as floor plans and elevations or sections, to construct more detailed 3D BIMs. However, these drawings are typically misaligned in orientation, scale, and projection dimensions, especially when derived from raster images. As a result, corresponding objects in different drawings cannot be directly matched without additional alignment or transformation processes. Moreover, most existing approaches rely heavily on CAD-based metadata, such as layer structures, block references [14], or textual annotations, to identify and associate architectural components across drawings. This dependence introduces structural limitations, making such methods vulnerable to inconsistencies in drawing conventions or the absence of metadata, which frequently occurs in real-world legacy architectural documents. Overcoming these limitations requires direct object matching between floor plans and elevations, yet no prior study has proposed a method for matching corresponding objects directly across misaligned raster drawings without relying on metadata. Accurate BIM creation requires the coordinated integration of information from both floor plans and elevations to eliminate representational gaps and achieve comprehensive geometric and semantic consistency within the hierarchical BIM data structure.

This study addresses these challenges through three primary contributions. First, it develops tailored dataset configurations and inference strategies for floor plans and elevations based on their distinct visual characteristics—tiled datasets with Slicing Aided Hyper Inference (SAHI) [15] for floor plans and combined datasets for elevations—achieving mAP@0.5 of 0.934 and 0.838, respectively, representing 41.8% and 16.2% improvements over baseline approaches. Second, it introduces an LMM-based reasoning framework for scale and dimensional calibration that enables dimensional information extraction from symbolic notations even under noisy or incomplete conditions, achieving 0.707 accuracy and substantially outperforming traditional OCR methods. Third, it establishes explicit object-level correspondence between floor plans and elevations through cost-matrix formulation and the Hungarian algorithm, achieving 0.93 correspondence F1-score, enabling comprehensive 3D BIM generation from multi-view drawings

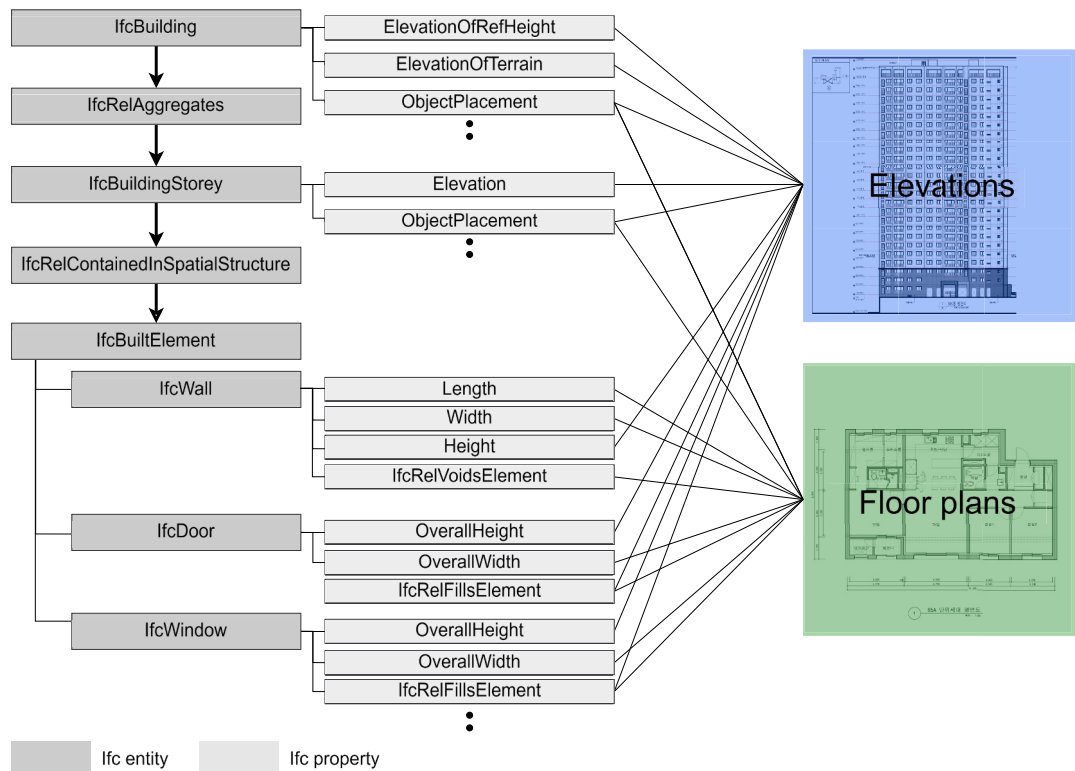


Fig. 1. Information represented in plans and elevations within the IFC-based hierarchical structure.

with incomplete or missing dimensional information.

The remainder of this paper is organized as follows: [Section 2](#) reviews related research and highlights the research gaps. [Section 3](#) describes the proposed method in detail. [Section 4](#) presents the evaluation procedure, [Section 5](#) reports the experimental results, and [Section 6](#) discusses the findings, implications, and limitations. [Section 7](#) concludes the paper.

2. Literature review

The automatic generation of 3D building models or BIM models from 2D architectural drawings has been extensively studied. Previous research can be broadly categorized into two groups according to the types of input drawings: (1) floor plan-based approaches and (2) multi-drawing-based approaches. This section critically reviews each category in turn and identifies the key research gaps that motivate this study.

2.1. Floor plan-based BIM generation approaches

Floor plans are widely used in building model generation research because they clearly represent the horizontal structure and spatial layout of buildings. They depict walls, openings, and rooms from a top-down perspective, enabling the extraction of object locations, adjacency relationships, and spatial configurations. Existing studies based on floor plans can be categorized into two groups: those that use CAD-based floor plans and those that use image-based floor plans.

CAD-based approaches often rely on CAD metadata, including layers and attribute tags defined within CAD files. For example, Byun and Sohn (2020) [16] proposed an automatic BIM generation system that extracted structural components such as columns and beams from structural floor plans and material schedules. Their method relied on structured CAD representations and standardized annotations to associate geometric elements with their corresponding attributes and generate parametric BIM elements. However, the applicability of the approach depended on the availability and consistency of such structured information. When CAD drawings deviated from predefined representation conventions or contained incomplete annotations, additional manual preprocessing or intervention was required to ensure reliable BIM generation. Similarly, Yang et al. (2020) [17] developed a semi-automated BIM generation process based on CAD construction drawings. They extracted object attributes from CAD layers and used grid-based relative positioning to determine locations. Their system allowed users to manually verify or modify extracted attributes. While this approach provided reliability, the system remained dependent on CAD layer information, making it vulnerable to missing, inconsistent, or ambiguous metadata. Similarly, Lim et al. (2018) [18] presented a semi-automated BIM generation method that processed multiple 2D CAD floor plans using parametric modeling tools. While notable time savings were achieved, the approach still depended on predefined or manually adjusted height values, highlighting the difficulty of fully automating vertical information extraction.

In contrast, image-based approaches rely on rule-based or deep learning methods to detect architectural objects. Gimenez et al. (2016) [11] applied image pattern matching to recognize wall elements and defined spaces based on topological relationships. After a graph was generated by combining wall connections and textual annotations, closed wall loops were used to define rooms, and 3D models were generated using fixed floor height values. However, the approach focused only on walls and rooms, excluding doors and windows. Additionally, all walls were assigned a uniform height, which limited the geometric precision of the resulting BIM. Subsequent studies introduced deep learning to improve recognition robustness under complex layouts.

Rho et al. (2021) [19] developed an algorithm to identify architectural objects by analyzing both line elements and text in floor plan images. Their method extracted vectorized geometry and examined intersections to define wall boundaries. Semantic attributes were added

by linking text annotations to geometric elements and were subsequently mapped to IFC-based BIM elements. However, the performance of the method was affected by the complexity of drawing layouts. Recognition accuracy declined when relationships between lines and text became ambiguous. Zhao et al. (2021) [12] employed a deep learning model using Faster R-CNN [20] to detect beams and columns in structural floor plans. Object positions and dimensions were determined using bounding boxes, and text annotations were extracted through OCR to define scaling parameters. These values were then used to convert pixel coordinates into real-world units and mapped to IFC elements. However, height values could not be extracted from the floor plan drawings alone, necessitating manual input of floor height and window height information.

Several studies further explored vectorization and 3D reconstruction from single floor plans. Liu et al. (2017) [21] proposed a learning-based raster-to-vector transformation framework that detected junctions and assembled architectural primitives to generate consistent vector layouts, enabling the creation of popup-style 3D models under predefined vertical assumptions. Park and Kim (2021) [22] introduced 3DPlanNet, which automatically generated 3D models from 2D floor plan images using a hybrid learning-based and rule-based framework, demonstrating the feasibility of large-scale 3D model generation without explicit dimension annotations. In addition, Kim et al. (2021) [23] proposed a patch-based deep learning framework that subdivided high-resolution floor plan images into fixed-size patches to normalize scale variations and improve recognition robustness, producing structured indoor spatial representations such as IndoorGML models.

In summary, floor plan-based approaches are effective in extracting horizontal geometry and spatial arrangements and have increasingly leveraged deep learning, vectorization, and patch-based processing to improve robustness. However, because most floor plans do not include explicit vertical information, height values are often predefined or manually specified. As a result, floor plan-only approaches remain inherently limited in capturing complete three-dimensional and parametric information, constraining the geometric accuracy and completeness required for fully automated BIM generation.

2.2. Multi drawing-based BIM generation approaches

To overcome the inherent limitations of floor plan-only approaches, particularly the absence of vertical information, several studies have explored BIM generation methods that integrate multiple types of architectural drawings. These approaches typically combine horizontal spatial layouts from floor plans with vertical information derived from elevation or section drawings to produce more complete three-dimensional building representations.

Early multi-drawing approaches predominantly relied on CAD drawings and their embedded metadata. Lu et al. (2020) [24] further investigated semi-automatic geometric digital twinning for existing buildings using 2D CAD drawings and related documentation. Their framework combined automated geometric processing with substantial manual intervention to generate three-dimensional building representations. Although the proposed method reduced modeling effort compared to fully manual workflows, it still relied on predefined modeling rules and user input, underscoring the persistent reliance on manual operations in multi-drawing-based BIM generation for existing buildings. Bortoluzzi et al. (2019) [14] presented an approach that integrated CAD-based floor plans and elevation drawings to infer the three-dimensional positions of windows and doors. Horizontal locations were extracted from floor plans, while vertical positions were derived from elevations using reference alignments. Semantic attributes were obtained from CAD layers and blocks before conversion into IFC-based BIM elements. While incorporating elevation drawings improved geometric completeness compared to floor plan-only methods, the approach relied on indirect coordinate alignment rather than explicit object-level correspondence and required well-structured CAD inputs.

Some studies further emphasized the use of elevation drawings to derive vertical information. Yin et al. (2020) [13] proposed a method that analyzed elevation drawings in CAD format to detect architectural elements such as windows, doors, and walls. Classification was based on CAD layer names, and floor heights were calculated from the relative positions between layers. This method successfully generated external building geometry in 3D. However, internal spatial components were not considered, and the approach was highly dependent on the accuracy and consistency of CAD metadata. In many real-world cases, layer information is missing or incorrect, making the method difficult to apply in practice.

In parallel with CAD-based multi-drawing workflows, several studies have explored the integration of multiple raster drawings using learning-based recognition techniques. Yetiş (2019) [25] proposed a semi-automated BIM generation workflow that integrated multiple raster architectural drawings, including floor plans and elevation views, to reconstruct three-dimensional building models. The method employed convolutional neural networks for semantic segmentation of architectural elements from raster images. Although the approach demonstrated the feasibility of multi-drawing integration without relying on CAD metadata, predefined geometric assumptions and rule-based reconstruction steps were still required for vertical modeling, limiting the level of full automation. Similarly, Urbieta et al. (2023) [26] proposed a framework that used both architectural and structural floor plans together with section drawings to generate 3D BIM elements. They applied Mask R-CNN [27] for instance segmentation and identified both architectural and structural components, including walls, windows, doors, beams, columns, and slabs. Rule-based geometric reasoning was used to integrate detected components across different drawing types, and scale normalization was performed to adjust coordinate systems. However, height information still had to be manually specified or derived from section views. Furthermore, scale symbols were not automatically detected and required manual confirmation.

Overall, multi-drawing-based approaches aim to address the limitations of individual drawing types by combining complementary information from multiple sources. Nevertheless, most existing methods rely on indirect coordinate alignment, predefined modeling rules, or extensive CAD metadata, and lack robust mechanisms for establishing explicit object-level correspondence between drawings. Moreover, significant manual intervention is often required, limiting their scalability and applicability to legacy raster drawings commonly encountered in practice.

2.3. Research gaps

Although automated BIM generation from 2D drawings has been actively explored, several critical gaps remain. These limitations can be summarized into four categories.

First, single-drawing approaches are inherently incomplete. Floor plan-based methods are effective in extracting horizontal layouts but cannot capture vertical information. Consequently, prior studies often assigned fixed or predefined height values, which reduced the geometric accuracy and realism of the generated BIMs. While drawings such as elevations or sections provide vertical information, methods relying on a single drawing type cannot represent both horizontal spatial configuration and vertical geometry in an integrated manner. As a result, single-drawing approaches remain insufficient for generating complete and geometrically consistent 3D BIMs.

Second, there exists a representation inconsistency between floor plans and elevations. Floor plans typically present balanced, top-down geometries, whereas elevation drawings present the vertical context of buildings through orthographic projection without perspective distortion. This inconsistency makes it difficult to apply a unified recognition or modeling strategy across both drawing types. Consequently, different dataset composition strategies (e.g., tiling for floor plans and combined processing for elevations) are often required, yet prior studies have not

provided a systematic framework for determining when and how such strategies should be applied.

Third, many existing scale calibration methods rely on explicit dimension annotations to convert pixel coordinates into metric values. Prior studies have shown that even when only a subset of dimension lines is correctly detected, robust scale estimation can be achieved through aggregation strategies such as majority voting across multiple annotations [28]. However, these approaches still fundamentally depend on the presence of dimension annotations and their numerical information. In real-world legacy drawings, dimension annotations may be partially missing, overlapped, or presented in heterogeneous formats, which can increase the uncertainty of scale inference and complicate fully automated processing. In contrast, symbolic scale indicators (e.g., “1:200” notations) are almost always present, but little research has investigated their systematic use for calibration. This dependency on dimension annotations continues to limit the robustness of automated BIM generation.

Finally, previous studies have not yet addressed direct object-level matching across floor plans and elevations. Some multi-drawing approaches have attempted to integrate information from different drawing types, but they generally rely on indirect coordinate alignment or relative positioning. None of these methods provides a robust mechanism for establishing explicit correspondence between objects in floor plans and elevations, particularly when dealing with misaligned raster drawings without CAD metadata. The absence of reliable cross-view object matching remains a critical barrier to achieving comprehensive BIM generation from 2D drawings.

To overcome these research gaps, this study aims to develop a method that directly matches architectural objects between floor plans and elevations using deep learning and spatial algorithms. Unlike prior approaches, the proposed method does not rely on CAD metadata and is designed to be robust to variations in drawing formats and styles. Furthermore, the proposed method incorporates a Large Multimodal Model (LMM) to extract scale information accurately, even from complex and unstructured drawings. A comparative summary of previous approaches and their limitations is presented in Table 1.

3. Proposed method

Fig. 2 presents the overall pipeline of the proposed method for automated BIM generation from floor plans and elevations. The proposed method is structured into two main phases, consisting of three and two steps respectively. The first phase, data interpretation and calibration from architectural drawings, comprises three sequential steps: (1) floor plan parsing, (2) elevation parsing, and (3) dimensional calibration using LMMs to extract scale data and convert pixel-based measurements into real-world units. The second phase, automated BIM generation, includes two steps: (1) multi-view correspondence matching between floor plans and elevations using a cost-matrix formulation and the Hungarian algorithm, (2) BIM element instantiation using refined spatial and semantic data obtained from the previous steps. This two-phase pipeline ensures consistent integration of geometric, semantic, and dimensional information extracted from 2D drawings into a coherent and information-rich BIM. The following subsections describe the proposed method in more detail. In this study, each floor plan is provided together with its corresponding floor index as an explicit input, rather than being inferred automatically. In addition, elevation drawings are processed with known directions (e.g., north, south, east, and west). Accordingly, the proposed method does not aim to identify which floor a given floor plan corresponds to. Instead, it focuses on establishing object-level correspondence between a given floor plan and the elevation-derived objects of the same floor.

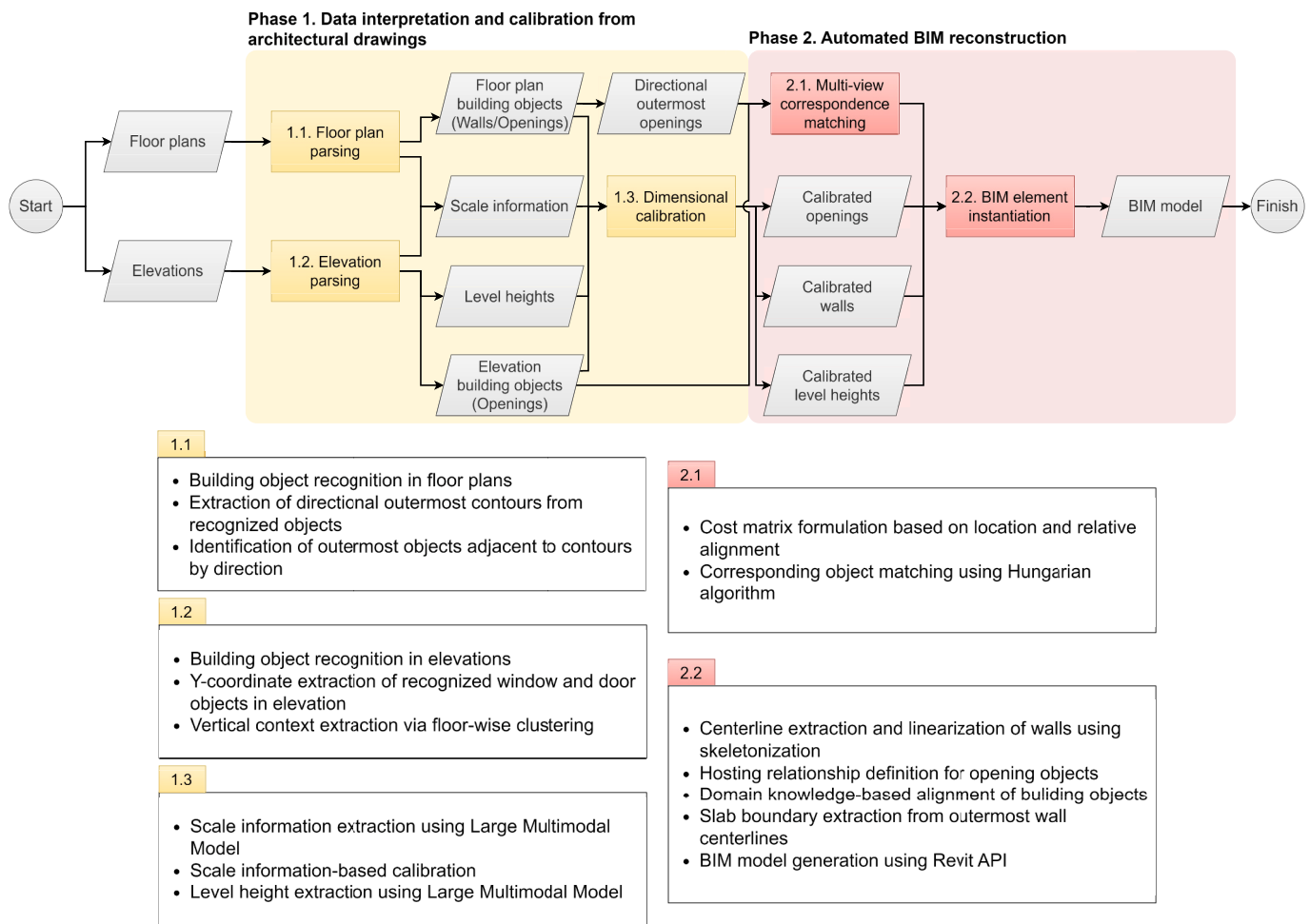
3.1. Data interpretation and calibration from architectural drawings

This section describes the process of interpreting architectural

Table 1

Comparison of information extraction and automation levels in previous studies on automated BIM generation.

Drawing type	Drawing format	References	Width/Length information	Height information	Dimensional information	Object correspondence
Floor plan	CAD	[16]	Automatically extracted	Manually set	Not addressed	Not addressed
		[17]	Automatically extracted	Pre-defined	Not addressed	Not addressed
		[18]	Automatically extracted	Pre-defined	Not addressed	Not addressed
	Image	[11]	Automatically extracted	Pre-defined	Not addressed	Not addressed
		[19]	Automatically extracted	Manually set	Not addressed	Not addressed
		[12]	Automatically extracted	Manually set	Automatically extracted	Not addressed
		[21]	Automatically extracted	Pre-defined	Not addressed	Not addressed
		[22]	Automatically extracted	Pre-defined	Automatically extracted	Not addressed
		[23]	Automatically extracted	Pre-defined	Automatically extracted	Not addressed
		[13]	Automatically extracted	Automatically extracted	Not addressed	Not addressed
Elevation	CAD	[26]	Automatically extracted	Automatically extracted	Automatically extracted	Not addressed
Floor plan + Section	Image	[26]	Automatically extracted	Automatically extracted or Manually set	Automatically extracted	Not addressed
Floor plan + Elevation	CAD	[14]	Automatically extracted	Automatically extracted	Not addressed	Semi-automated
		[24]	Automatically extracted	Pre-defined	Not addressed	Not addressed
	Image	[25]	Automatically extracted	Pre-defined	Automatically extracted	Not addressed
		Ours	Automatically extracted	Automatically extracted	Automatically extracted	Automatically extracted

**Fig. 2.** Proposed method pipeline.

drawings and extracting structured information for BIM generation. The proposed method consists of three steps: floor plan parsing and elevation parsing to extract horizontal and vertical spatial information, and dimensional calibration to convert object dimensions into real-world metric units.

3.1.1. Floor plan parsing

This step aims to extract horizontal spatial information of

architectural objects from floor plans and prepare them for integration with elevation data for 3D BIM generation. To this end, small and densely packed objects in large-scale floor plan images were recognized using a combination of the trained YOLO11s-seg model [29] and SAHI [15]. Based on the resulting segmentation masks, the slab boundary of each building was defined using morphological processing and contour detection techniques. Window and door objects located near the slab boundary were then identified as outermost elements, and their

corresponding elevation directions were determined by analyzing spatial relationships (Fig. 3).

The trained YOLO11s-seg model was used to detect architectural objects. Due to the high density of small objects typically present in large-scale architectural drawings, conventional YOLO models often exhibit decreased performance in detecting small-scale features. To address this limitation, SAHI was applied to enhance detection performance for small objects within high-resolution drawings. SAHI operates in a manner similar to the traditional sliding window approach in computer vision tasks. It divides the input image into fixed-size tiles, performs object detection on each tile independently, and then merges the detection results back into the original image coordinate system. A 20% overlap was applied between adjacent tiles to prevent the loss of objects located at tile boundaries.

The recognition results were saved in JSON format and included information such as class labels, segmentation masks, bounding boxes, and the center coordinates of each bounding box. Following object recognition, the slab was defined based on the resulting segmentation masks. First, the masks of detected objects were binarized to generate a binary mask image of the entire floor plan. To address small gaps and discontinuities along object boundaries, morphological operations (e.g., dilation and closing) were applied sequentially using OpenCV. This preprocessing step enabled adjacent or partially disconnected objects to be merged into a single coherent structure. After preprocessing, a contour detection algorithm was applied to the binary mask. Among the detected contours, the largest closed shape in terms of area was selected as the outermost contour.

To enable matching between floor plans and elevations, window and door objects located near the outermost contour were identified as outermost objects. The classification of outermost objects was determined by calculating the minimum distance between the center point of each object and the slab boundary. If the distance was below a predefined threshold, the object was labeled as outermost. To determine the corresponding elevation direction of each outermost object, reference points were extracted from the top, bottom, left, and right edges of the slab boundary. Euclidean distances were then calculated between each object's center and the reference points. The initial direction was assigned based on the closest reference point, and the final direction was

determined using a majority voting scheme that considered the distance distribution between all boundary coordinates of the object and each reference region. Through this process, each object was assigned two attributes: whether it is classified as an outermost object and its corresponding elevation direction.

3.1.2. Elevation parsing

This step focuses on extracting vertical spatial information from elevation drawings and integrating it with the horizontal coordinate data obtained from floor plans to create the 3D geometry of architectural objects for more accurate BIM generation. The process begins with detecting key architectural objects in elevations using the YOLO11s-seg instance segmentation model, followed by clustering the centroids of detected objects to estimate floor-level information. To enable reliable matching between objects in floor plans and corresponding elements in each elevation, a directional outermost filtering strategy was applied. The directional outermost objects refer to the boundary-side elements located along the outermost edges of the floor plan facing a specific direction (e.g., South, East, North, or West). These objects define the visible façade for each elevation and thus serve as the key linkage between objects of floor plans and elevations. Finally, the detected objects are aligned to the real-world coordinate system of the building. The elevation recognition process is illustrated in Fig. 4.

To improve object recognition performance on high-resolution elevation images, the trained model was combined with SAHI. Since wall objects are typically represented as large components in elevations, Full Inference (FI) was applied to detect wall elements across the entire image. Similar to the floor plan recognition process, SAHI was used to improve detection of small components in large-scale images by dividing the drawing into overlapping tiles. A 20% overlap between tiles was also applied to minimize the risk of missing objects at tile boundaries.

The detection results were stored in JSON format, including class labels, segmentation masks, bounding boxes, and the center coordinates of each bounding box. Since the coordinate basis varies depending on the orientation of the elevation (e.g., South, East, North, or West), object coordinates were converted accordingly to match the building's spatial reference system. For South- and North-facing elevations, the x-axis was used as the basis for horizontal alignment, whereas the y-axis was used

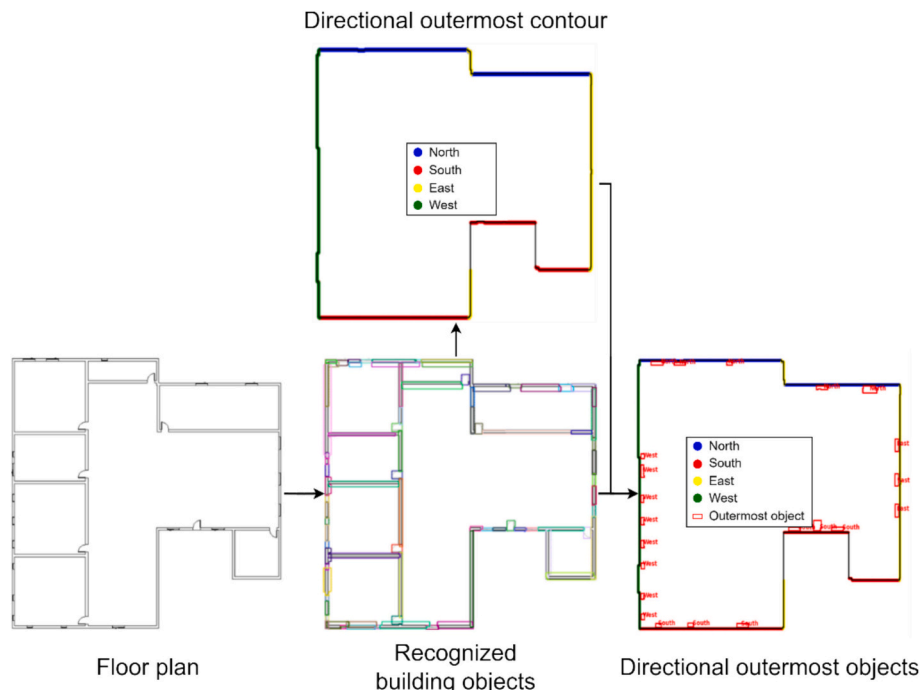


Fig. 3. Floor plan parsing process.

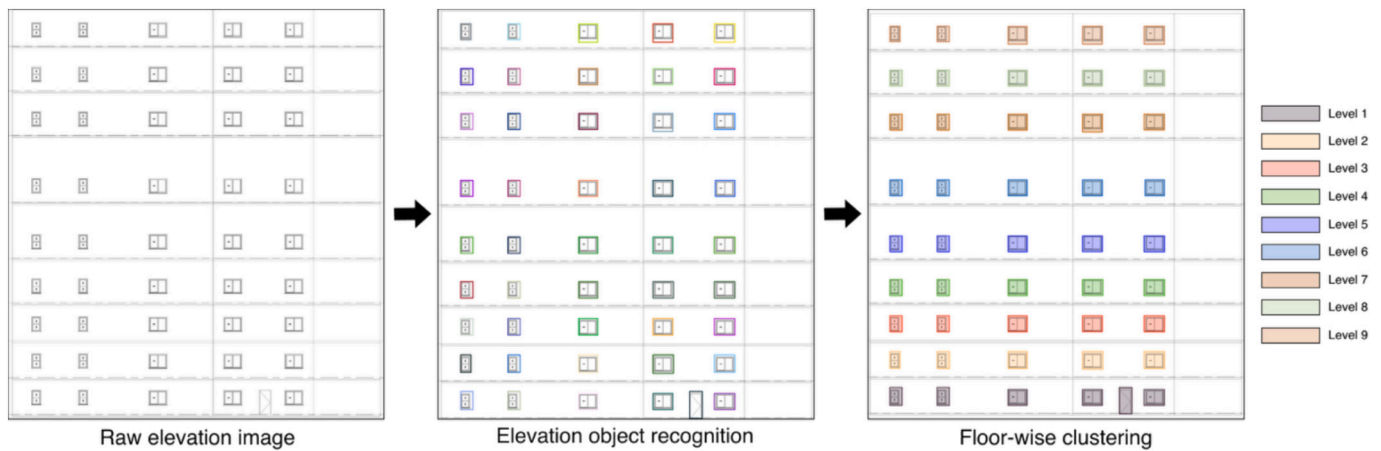


Fig. 4. Elevation parsing process.

for East- and West-facing elevations. The transformed coordinates were subsequently used as input for matching algorithms to identify corresponding objects between elevation and floor plan drawings.

The center coordinates of window and door objects detected in elevation drawings played a key role in estimating floor-level information. To classify the objects by floor, clustering was performed using the y-axis values of the transformed center coordinates, which represent the vertical direction in the building. The clustering algorithm used was Density-Based Spatial Clustering of Applications with Noise (DBSCAN), which groups data points based on density and treats sparse points that do not meet the density threshold (ϵ) as noise [30]. Unlike simple threshold-based segmentation, floor-level separation in elevation drawings cannot be reliably achieved using fixed vertical intervals due to irregular floor spacing and architectural variations. Moreover, floor heights vary across buildings and even across different floors within the same building, and a representative or maximum floor height is generally unknown a priori from elevation drawings. Similarly, clustering methods such as k-means require the number of clusters to be predefined, which is impractical when the number of floors is unknown a priori. DBSCAN was therefore adopted to robustly identify floor-level clusters without requiring prior input of the number of floors, while effectively handling detection noise and outliers commonly observed in real-world elevation drawings. In this study, ϵ was set to 50. Each object was assigned a floor number based on the average y-coordinate of its cluster.

The coordinate values of the elevation objects were originally represented in the image's pixel coordinate system. However, the pixel coordinate system defines the top of the image as $y = 0$, whereas the real-world coordinate system defines the bottom of the building as $y = 0$. Therefore, the coordinates needed to be converted to match the world coordinate system used in BIM generation. The highest-positioned wall object in the image (i.e., the one with the smallest y-value in pixel coordinates) was used as a reference point. All y-coordinates were then redefined as the relative vertical distances from this reference point, effectively transforming them to align with the actual vertical orientation of the building.

3.1.3. Dimensional calibration of building objects

This step introduces a method for recognizing the scale information embedded in architectural drawings and converting pixel-based coordinates into real-world metric units. Since instance segmentation outputs provide object geometries in image space, applying an accurate scale transformation is essential for generating valid BIMs that reflect real-world dimensions.

However, converting pixel coordinates into metric values typically requires reliable recognition of dimension lines or textual annotations

that specify scale information. In real-world architectural drawings, such annotations are often presented in inconsistent formats, written in varying styles and positions, or are partially missing due to overlaps or drawing artifacts. These limitations make it difficult for conventional OCR methods to extract scale values with high accuracy. As a result, many traditional methods fail to ensure consistent unit conversions, leading to significant errors in object dimensions.

To address these limitations, this study employs an LMM to interpret scale annotations based on both visual layout and textual content. An overview of the LMM-based scale recognition process is illustrated in Fig. 5. By leveraging the model's ability to reason across image and text modalities, scale values can be extracted more reliably even when presented in non-standard formats or partially occluded conditions. Since the input is a digital image, the image resolution in pixels is directly available. The physical scale of the drawing is obtained by reading scale annotations embedded in the drawing using the LMM. The extracted scale factor is then used to convert all object positions and sizes from pixel units into real-world metric units. This approach is applicable to drawings in which scale annotations are present, a common characteristic of architectural drawings used for construction and documentation.

Each input drawing is first divided into fixed-size tiles of 1000×1000 pixels, and each tile is independently analyzed using the GPT-4o model. The GPT-4o version used in this study corresponds to the 2024-11-20 release. A fixed decoding configuration is adopted, with the temperature set to 0, top-p set to 1.0, and the maximum number of generated tokens limited to 32. During inference, the model searches for scale-related keywords such as "SCALE", "1/100", and "S = 1:200". The detected results are normalized into a predefined format (e.g., <paper size>: <scale>, such as A1: 1/100) through a structured prompt that constrains the model output to a single scale value without additional explanations.

The extracted outputs are subsequently validated using rule-based pattern checks to ensure compliance with the predefined format. Outputs that do not satisfy the format constraints are reprocessed, while tiles that do not contain valid scale information are discarded. When multiple scale annotations are detected across different tiles of the same drawing, they typically represent semantically equivalent scale specifications expressed with respect to different paper formats (e.g., A1: 1/200 and A3: 1/400). In such cases, a deterministic selection rule is applied to retain a single representative value by prioritizing standard paper sizes (A1 first, otherwise A3). This procedure ensures consistent and reproducible scale extraction without manual intervention.

Once the final scale value is determined, pixel-to-metric conversion is performed using the image resolution in pixels and the known physical dimensions of the corresponding paper size. The pixel-to-metric conversion factor is computed as follows:

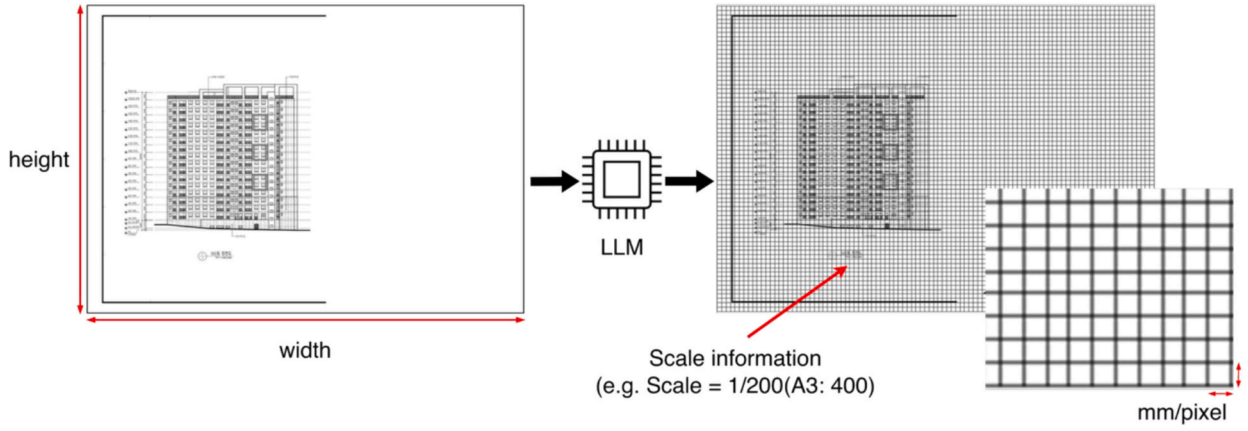


Fig. 5. LMM-based scale recognition.

$$\text{Conversion Factor (mm per pixel)} = \frac{\text{Paper width(mm)}}{\text{Image width(px)}} \times \text{Scale factor} \quad (1)$$

For example, if a 5000-pixel-wide image represents an A1 drawing (841 mm wide) at a scale of 1:100, the resulting conversion factor becomes:

$$\text{Conversion Factor} = \frac{841(\text{mm})}{5000(\text{px})} \times 100 = 16.82 \text{ mm/px} \quad (2)$$

In addition to scale calibration, floor height information is extracted using the same LMM-based framework to determine the vertical positions of architectural objects. Each elevation drawing is encoded and processed by GPT-4o with a structured prompt enforcing JSON-formatted output that returns individual floor heights and the total building height. The extracted height values are mapped to objects based on their assigned floor levels obtained from clustering results. When height information is unavailable or extraction fails, default values of 1200 mm for windows and 2100 mm for doors are applied. This calibration ensures that all object coordinates are metrically consistent, establishing a unified spatial basis for 3D BIM generation.

3.2. Automated BIM generation

This section describes the automated BIM generation process based on the interpreted drawing information obtained from Section 3.1. The proposed method consists of two main steps: multi-view correspondence matching, which establishes explicit object-level correspondences between floor plans and elevations to derive complete 3D coordinates, and BIM element instantiation, which generates a geometrically consistent 3D BIM model by refining wall centerlines and defining hosting relationships between walls and opening elements.

3.2.1. Multi-view correspondence matching

This step proposes a matching method that automatically identifies correspondences between architectural objects recognized in floor plans and elevation drawings, which are then used to determine the complete 3D coordinates for each object. While floor plans provide horizontal positional information, elevation drawings contain vertical position and directional data. By aligning corresponding objects across both types of drawings, it becomes possible to generate the full 3D spatial configuration. To achieve this, it is essential to develop a method that accurately matches the same object represented in both the floor plan and the elevation. Previous studies have relied on CAD layer information to align objects based on common reference points, followed by indirect attribute-based matching using relative positional comparisons. However, such approaches struggle to establish explicit one-to-one corre-

spondences and depend heavily on the presence of indirect attribute information within the drawings. As a result, the matching accuracy tends to degrade when the necessary information is missing or imprecise.

Algorithm 1. Direction-normalized corresponding object matching between floor plans and elevations.

Require :

- Detected floor plan objects O_i^{FP} with center coordinates $Center(O_i^{FP}) = (x_{FP,i}, y_{FP,i})$, and image dimensions (W_{FP}, H_{FP}) ;
- Detected elevation objects O_j^{ELEV} with center coordinates $Center(O_j^{ELEV}) = (u_{ELEV,j}, v_{ELEV,j})$, and image dimensions (W_{ELEV}, H_{ELEV}) , and drawing directions $dir_j \in \{\text{South, East, North, West}\}$

Ensure : Matched object pairs (O_i^{FP}, O_j^{ELEV}) with 3D coordinates $(x_{FP,i}, y_{FP,i}, z_{ELEV,j})$.

```

1: Step 1: Coordinate conversion for elevation objects
2: for  $j \leftarrow 1$  to  $M$  do
3:   if  $dir_j \in \{\text{South, East}\}$  then
4:      $x_{ELEV,j} \leftarrow u_{ELEV,j}$ 
5:   else if  $dir_j \in \{\text{North, West}\}$  then
6:      $x_{ELEV,j} \leftarrow W_{ELEV} - u_{ELEV,j}$ 
7:   end if
8:    $z_{ELEV,j} \leftarrow H_{ELEV} - v_{ELEV,j}$ 
9: end for
10: Step 2: Coordinate normalization (scale-invariant matching)
11: for  $i \leftarrow 1$  to  $N$  do
12:    $x_{FP,i}^* \leftarrow x_{FP,i} / W_{FP}$ 
13:    $y_{FP,i}^* \leftarrow y_{FP,i} / H_{FP}$ 
14: end for
15: for  $j \leftarrow 1$  to  $M$  do
16:    $x_{ELEV,j}^* \leftarrow x_{ELEV,j} / W_{ELEV}$ 
17: end for
18: Step 3: Cost matrix computation
19: Initialize cost matrix  $C \in \mathbb{R}^{N \times M}$ 
20: for  $i \leftarrow 1$  to  $N$  do
21:   for  $j \leftarrow 1$  to  $M$  do
22:     if  $dir_j \in \{\text{South, North}\}$  then
23:        $C_{ij} \leftarrow |x_{FP,i}^* - x_{ELEV,j}^*|$ 
24:     else if  $dir_j \in \{\text{East, West}\}$  then
25:        $C_{ij} \leftarrow |y_{FP,i}^* - y_{ELEV,j}^*|$ 
26:     end if
27:   end for
28: end for
29: Step 4: Hungarian matching and 3D coordinate assignment
30: if  $N \neq M$  then
31:   Add dummy rows or columns to  $C$  with large penalty values
32: end if
33:  $(rows, cols) \leftarrow \text{Hungarian}(C)$ 
34: for all  $(i, j)$  in  $(rows, cols)$  do
35:   if  $j$  is not dummy then
36:     Assign matched pair  $(O_i^{FP}, O_j^{ELEV})$ 

```

(continued on next page)

(continued)

Require :

```

37:   Compute 3D coordinate  $(x_{FP,i}, y_{FP,i}, z_{ELEV,j})$ 
38:   end if
39: end for
40: return Matched pairs and 3D coordinates

```

To address these limitations, this study introduces a matching method based on the center coordinates of the recognized objects. A cost matrix was constructed by quantifying similarity based on coordinate differences, and the Hungarian algorithm [31] was employed to perform optimal one-to-one matching. The Hungarian algorithm minimizes the total matching cost, allowing for stable matching even when the number of detected objects differs across the two drawings or when the coordinate information is not perfectly aligned. The detailed steps of this matching process are described in Algorithm 1.

The matching process was based on the positional differences between the centers of detected objects in the floor plans and elevations. Each floor plan object O_i^{FP} and elevation object O_j^{ELEV} was represented by its center coordinate as

$$Center(O_i^{FP}) = (x_{FP,i}, y_{FP,i}), Center(O_j^{ELEV}) = (u_{ELEV,j}, v_{ELEV,j}). \quad (3)$$

The centroid of floor plan objects $(x_{FP,i}, y_{FP,i})$, was extracted from the segmentation masks. For elevation drawings, the detected object centroid $(u_{ELEV,j}, v_{ELEV,j})$ was transformed to a unified south-facing reference based on the drawing direction $dir_j \in \{South, East, North, West\}$:

$$x_{ELEV,j} = \begin{cases} u_{ELEV,j}, dir_j \in \{South, East\} \\ W_{ELEV} - u_{ELEV,j}, dir_j \in \{North, West\} \end{cases}, \quad (4)$$

$$z_{ELEV,j} = H_{ELEV} - v_{ELEV,j}. \quad (5)$$

Here, W_{ELEV} and H_{ELEV} denote the elevation image width and height in pixels, respectively. The horizontal mirroring by $W_{ELEV} - u_{ELEV,j}$ for north- and west-facing elevations ensures that all elevations are aligned to a consistent south-facing reference frame, while the vertical inversion by $H_{ELEV} - v_{ELEV,j}$ defines the upward direction of the height axis z .

Because floor plan and elevation images may have different resolutions, all coordinates were normalized by their image dimensions prior to cost computation:

$$x_{FP,i}^* = \frac{x_{FP,i}}{W_{FP}}, y_{FP,i}^* = \frac{y_{FP,i}}{W_{FP}}, x_{ELEV,j}^* = \frac{x_{ELEV,j}}{W_{ELEV}}. \quad (6)$$

The pairwise matching cost between O_i^{FP} and O_j^{ELEV} was then defined as the absolute difference of their normalized horizontal positions:

$$C_{ij} = \begin{cases} |x_{FP,i}^* - x_{ELEV,j}^*|, dir_j \in \{South, North\} \\ |y_{FP,i}^* - x_{ELEV,j}^*|, dir_j \in \{East, West\} \end{cases} \quad (7)$$

The cost matrix C_{ij} represents the horizontal positional discrepancy between each floor plan object and elevation object, serving as the basis for computing the optimal matching through the Hungarian algorithm. The Hungarian algorithm was applied to the cost matrix $C \in \mathbb{R}^{N \times M}$ to obtain the optimal one-to-one assignment between the N detected floor plan objects and M elevation objects.

When $N \neq M$, dummy rows or columns were added to form a square matrix, and large penalty values were assigned to dummy pairs to prioritize real matches. After the optimal assignment was determined, the vertical coordinate $z_{ELEV,j}$ of each matched elevation object (in non-normalized pixel units) was appended to its corresponding floor plan object to form the complete 3D coordinate:

$$(x_{FP,i}, y_{FP,i}, z_{ELEV,j}). \quad (8)$$

This process ensured consistent directional alignment between floor plan and elevation data. As illustrated in Fig. 6, the integration of horizontal coordinates (x, y) from floor plans and vertical coordinates (z) from elevations enables accurate generation of wall, window, and door objects.

3.2.2. BIM element instantiation

This step performs the automatic generation of BIM by integrating spatial and semantic data derived from object recognition. To support accurate modeling, the recognized object data are first post-processed to extract simplified wall centerlines and define hosting relationships between walls and opening elements. These processes ensure that architectural components are geometrically refined and spatially aligned.

1) Centerline extraction and linearization of walls using skeletonization

Each wall object recognized from the floor plan contains complex polygonal contours, which are unsuitable for direct BIM generation. A Manhattan-world assumption [32] is adopted, assuming that building geometry is predominantly orthogonal, to enable the representation of wall centerlines as vertical or horizontal segments. To simplify the geometry, skeletonization is applied to each wall mask using OpenCV's thinning algorithm, yielding centerline coordinates of the wall segments. However, these raw skeleton outputs often include noise, such as subtle deviations in coordinates near curves, which can lead to inaccurate corner point definitions. An example of this post-processing procedure is illustrated in Fig. 7.

To mitigate this, a grid-based snapping operation is applied to all skeleton coordinates, removing minor deviations and merging adjacent points into cleaner, unified coordinates. In this study, a fixed grid size of 30 pixels is used, where the value was empirically determined based on the image resolution and the typical thickness of wall segments in the drawings. This pixel-level grid was found to be sufficient to suppress noise introduced during skeletonization while preserving the overall geometric structure. The refined skeleton points are then grouped into linear segments based on shared x - or y -values, classifying them into either vertical or horizontal lines. Each line is defined by a start point and an end point, which serve as corner points for further processing. This procedure enables efficient and accurate extraction of centerlines from otherwise complex wall segmentations.

2) Defining hosting relationships for opening objects

To ensure accurate BIM generation, structural relationships between wall and opening objects (windows and doors) must be explicitly defined. Each opening must be assigned to a specific wall where it is physically located. This is achieved by establishing a hosting relationship for each window and door object. Fig. 8 illustrates examples of how these spatial relationships are defined for both window and door objects.

For windows, the relative orientation (horizontal or vertical) is determined by comparing the width and height of the bounding box. The shorter side is assumed to be in contact with a wall, and the window is classified accordingly. The algorithm then searches for the nearest wall segment in the same orientation and adjusts the window's center position to project onto that segment. The identified wall is conceptually assigned as the host of the window, ensuring that its spatial relationship is accurately reflected in the BIM.

For doors, the association with walls is first checked by examining whether the bounding box intersects with any wall centerlines. If an intersection is found, the center of the door is adjusted to the intersection point, and the corresponding wall is designated as its host. If there is no intersection, the nearest wall is identified, and the door's center is projected onto this wall segment, following the same procedure used for windows. By defining these hosting relationships, window and door objects are positioned with greater precision relative to their corresponding walls.

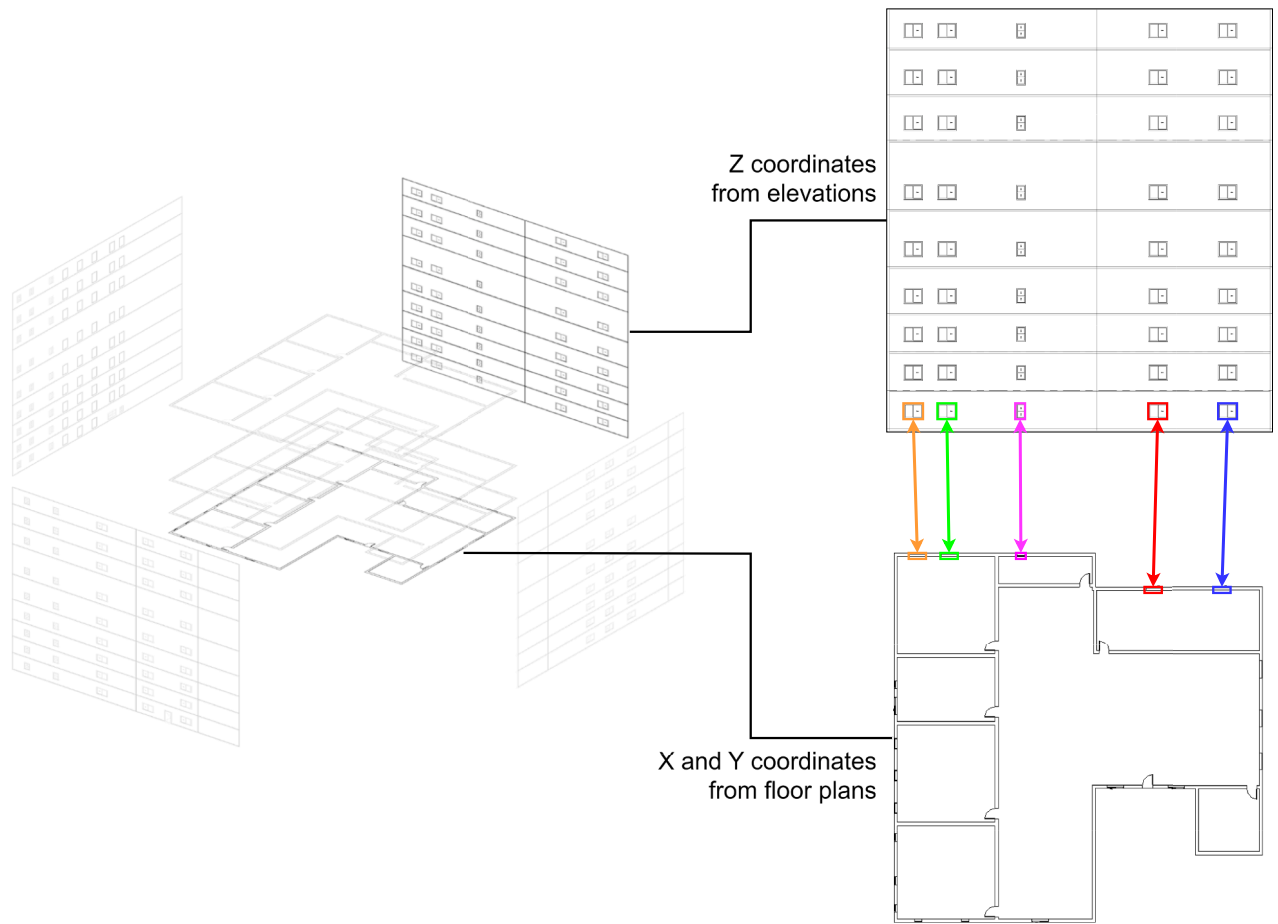


Fig. 6. Corresponding object matching between floor plans and elevations.

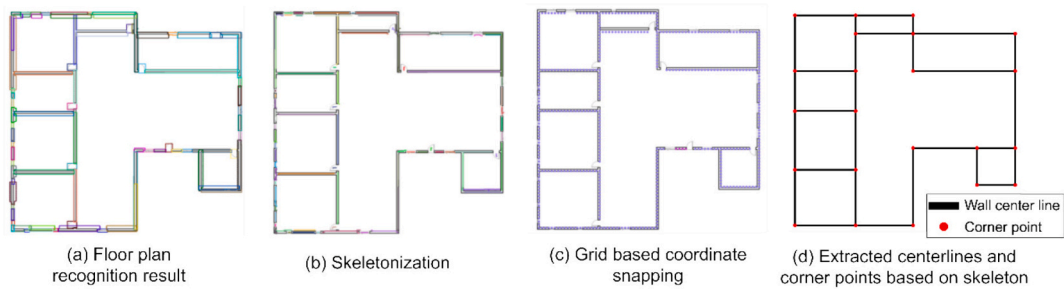


Fig. 7. Post-processing for wall centerline extraction and linearization.

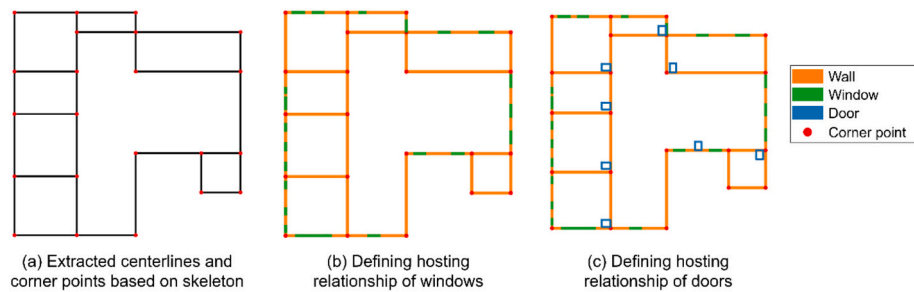


Fig. 8. Defining spatial relationships for opening objects.

3) Domain knowledge-based adjustment of building objects

To improve consistency of object placement across floors, a domain knowledge-based alignment strategy was applied to the recognized

floor plan objects. Although instance segmentation models effectively detect objects such as windows and doors, their outputs often exhibit small positional or dimensional inconsistencies across floors due to pixel-level variances and edge segmentation errors. These discrepancies become critical when identical architectural elements (e.g., windows) are repeated on multiple floors but appear with slightly different coordinates or sizes.

The alignment procedure was implemented in four steps. First, objects of the same type (window or door) were grouped floor-wise based on their (x, y) positions, sizes (width, height), and orientations using clustering. Second, one representative floor, typically the first floor with the most complete layout, was selected as the *anchor*. Objects in this floor served as reference instances for aligning corresponding elements on other floors. Third, object pairs were matched across floors if their spatial and dimensional differences fell within tolerance thresholds: ± 20 mm for position and ± 30 mm for size. These thresholds reflect acceptable deviations in practice and are sufficiently small to correct segmentation-induced noise without collapsing distinct elements into a single instance. Finally, for matched pairs, the coordinates and dimensions of non-anchor objects were adjusted to exactly match those of the anchor, ensuring consistent alignment across the entire building. This rule-based correction significantly reduced inter-floor inconsistencies, enhanced the geometric regularity of the BIM output, and better reflected the design logic commonly followed in multi-story residential buildings.

The model generation process involved defining building levels from extracted floor height data, instantiating wall and opening objects using their geometric parameters, and applying hosting relationships between windows, doors, and walls. In addition, slab boundaries were defined by tracing the outermost wall centerlines derived from the refined geometric data, and these closed contours were converted into planar floor elements to represent each story's base structure within the BIM environment.

Each object was placed according to its center coordinates, dimensions, and assigned level. Hosting information ensured that opening objects were embedded into their respective walls. The model generation routine reused predefined Revit FamilySymbol components when applicable, enabling consistent and efficient creation of identical elements across multiple stories. Through this automated procedure, a complete BIM was generated within the Revit environment, including correctly aligned slabs, walls, windows, and doors along with their spatial relationships (Fig. 9).

4. Experimental design

This section presents the dataset, experimental setup, and metrics used to validate the effectiveness of the proposed method. The evaluation assesses both the individual steps of the proposed method and the final BIM output, ensuring a comprehensive analysis of system performance. The experiment was designed to systematically verify the

effectiveness of the proposed method across its sequential steps. Since each step in the pipeline produces outputs that serve as inputs for the subsequent one, the validation was conducted in a progressive manner to ensure reliability throughout the workflow. The entire experiment is structured into two major evaluation groups. The first group, drawing interpretation, comprises three steps: architectural object recognition, floor-wise clustering, and scale information recognition. The second group, automated BIM generation, comprises two steps: cross-view correspondence matching and BIM element generation. An overview of the experimental design is illustrated in Fig. 10.

4.1. Evaluation of drawing interpretation

This section evaluates the performance of the drawing interpretation process, including architectural object recognition, floor-wise clustering, and scale information recognition, to assess the reliability of extracting spatial and dimensional information from architectural drawings.

4.1.1. Architectural drawing object recognition

The objective of this step was to evaluate how effectively the trained models could identify walls, windows, and doors under different dataset configurations and inference strategies.

This study utilized the “Architectural Drawing Dataset” provided by AI-Hub [33], South Korea. The dataset primarily consists of floor plans and elevations of buildings. A key feature of this dataset is that object-level annotations are provided in COCO JSON format for each drawing image. In this study, the object classes were limited to walls, windows, and doors, and the training dataset was constructed based on these categories. The label of the dataset was converted into a format suitable for instance segmentation models. Since the size of each drawing image is 4963×3509 pixels, the image tiling technique was applied to address limitations in computational resources. Image tiling is a data preprocessing strategy that divides large images into smaller patches, enabling object detection models to train more efficiently [34]. However, when objects are located on the edges of tiles, they may be split, resulting in missing objects or loss of label information during training [35]. To mitigate this issue and prevent performance degradation, a 10% overlap was applied between adjacent tiles.

Three types of datasets were constructed (Fig. 11). First, the original dataset was built using unprocessed full-sized drawing images, allowing the model to learn contextual information from large-scale images. Second, the tiled dataset was created by dividing floor plan images into 1280×1280 pixel patches and elevation images into 1024×1024 pixel patches, facilitating the accurate recognition of relatively small-scale objects. Third, the combined dataset included both original and tiled images, enabling the model to learn both global and local features simultaneously. Images without any annotated objects were excluded during preprocessing to improve training efficiency and accuracy. For floor plan object recognition model training, the original dataset

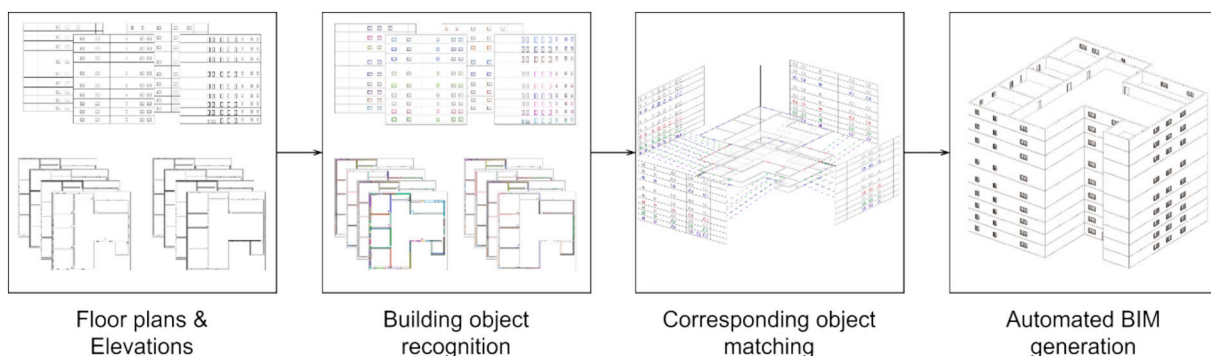


Fig. 9. BIM generation procedure.

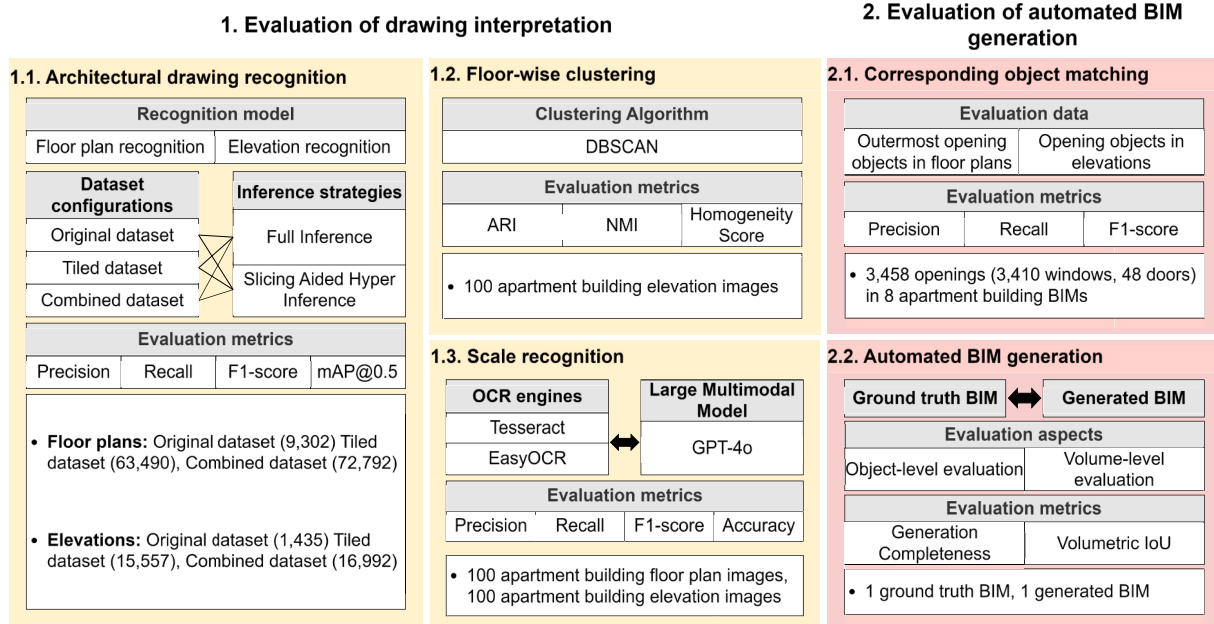


Fig. 10. Experimental design.

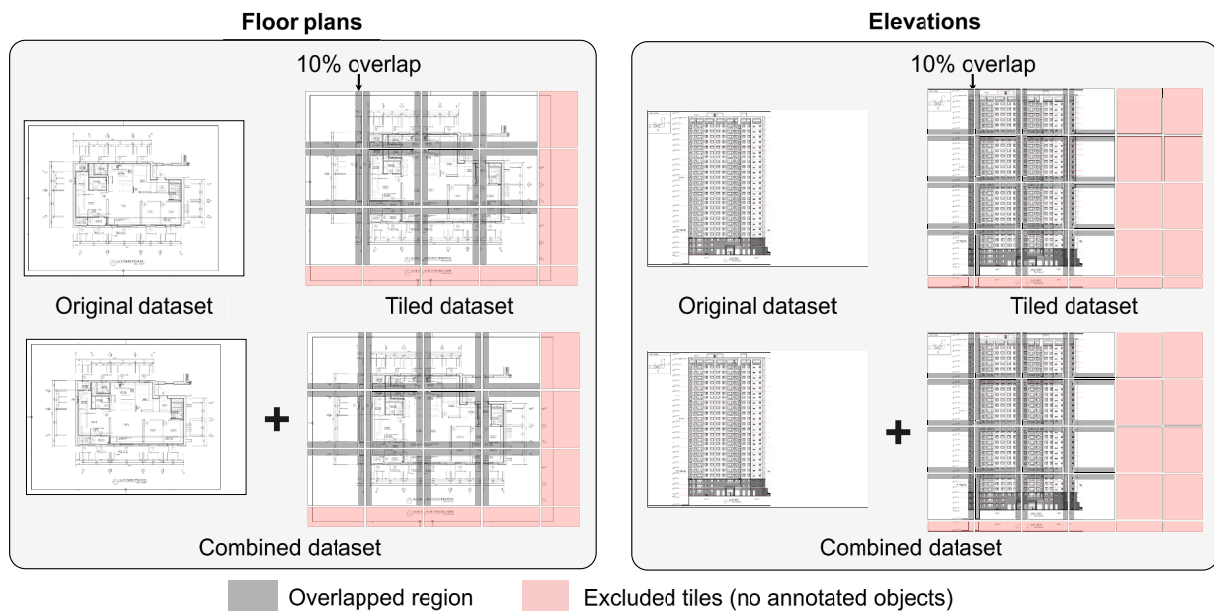


Fig. 11. Examples of original, tiled, and combined training datasets.

consisted of 7442 training images and 1860 validation images, while the tiled dataset contained 50,792 training images and 12,698 validation images. The combined dataset, formed by merging the two, included 58,234 training images and 14,558 validation images. For elevation object recognition model training, the original dataset consisted of 1147 training images and 288 validation images, while the tiled dataset included 12,445 training images and 3112 validation images. The combined dataset contained 13,592 training images and 3400 validation images. The number of images in each dataset type is summarized in Table 2.

To automatically recognize architectural objects in floor plans, an instance segmentation model based on the YOLO (You Only Look Once) architecture [36] was employed. YOLO is one of the most representative one-stage detector models that simultaneously predicts object locations

Table 2
Dataset configuration.

Drawing type	Configuration types (total/train/validation)		
	Original	Tiled	Combined
Floor plans	7442 / 1860	50,792 / 12,698	58,234 / 14,558
Elevations	1147 / 288	12,445 / 3112	13,592 / 3400

and classes through a single neural network. It is known for its fast inference speed and high accuracy. In this study, the YOLO11-seg architecture, a recent instance segmentation version of YOLO, was used as the base model. Considering the limitations of available GPU resources, the lightweight version, YOLO11s-seg [29], was selected for training. Model training was conducted independently for each of the three

dataset types (i.e., original, tiled, and combined), and the model that achieved the best performance was selected as the final floor plan recognition model. To address the issue of class imbalance, the Focal Loss function was applied during training [37]. The optimizer used was Stochastic Gradient Descent (SGD), which is a classical optimization technique [38]. Since the floor plan dataset exhibited relatively uniform object distributions and stable convergence behavior, SGD was adopted to ensure training stability. Different weight decay values were applied to different parameter groups: a decay of 0.0005 was applied to weights, while no decay was applied to biases and normalization parameters. The learning rate was set to 0.01, momentum to 0.937, batch size to 16, and the input image size to 1280×1280 pixels. The model was trained for 100 epochs.

The elevation object recognition model was also trained using the YOLO11s-seg instance segmentation architecture. The object classes (i.e., walls, windows, and doors) were set identically to those used in the floor plan recognition model. The Adaptive Moment Estimation with decoupled Weight Decay (AdamW) optimizer was adopted [39]. Given the highly imbalanced distribution of objects in the elevation dataset, AdamW was deemed appropriate to facilitate faster convergence during the early stages of training. The input image size was set to 1024×1024 pixels, while the remaining hyperparameters were configured identically to those used in the floor plan model. The final set of hyperparameter configurations used for model training is summarized in Table 3.

To evaluate the recognition performance of the trained models, comparative experiments were conducted under different inference strategies and dataset configurations. Specifically, two inference strategies were tested: FI, which processes entire drawings at once, and SAHI, which performs localized inference on overlapping tiles and merges the results into the original coordinate space [15]. Each model was evaluated across three dataset configurations (Original, Tiled, and Combined) and two inference modes (FI and SAHI), resulting in six experimental conditions per drawing type.

4.1.2. Floor-wise clustering

The second step focused on floor-wise clustering of elevation objects. Detected windows and doors were clustered based on their vertical positions to identify building stories. The clustering performance was evaluated to confirm whether the method could correctly distinguish floor levels even in multi-story buildings with irregular facade layouts. Accurate identification of story boundaries was essential for linking elevation data with corresponding plan information. For the floor-wise clustering step, elevation drawings were collected from 100 apartment building elevations. The vertical centroid coordinates of window and door objects detected by the trained model were used as inputs for density-based clustering. The clustering results were compared with manually labeled floor-level ground-truth to evaluate the method's ability to identify distinct stories.

4.1.3. Scale information recognition

The third step tested the performance of scale information recognition, which enables conversion of pixel-based coordinates into real-world dimensions. This step compares the recognition accuracy of

LMMs with conventional OCR-based methods, focusing on their ability to interpret scale annotations in various formats. Reliable scale recognition was critical for ensuring metric accuracy in the final BIM output. For this step, 100 elevations and 100 floor plans were used to test both the detection and recognition of scale annotations. These drawings were sampled from the same drawing dataset and represented diverse notational styles and layout complexities. The results from LMMs and conventional OCR engines were compared against manually verified ground-truth values.

4.2. Evaluation of automated BIM generation

This section evaluates the performance of automated BIM generation, including corresponding object matching between floor plans and elevations and the quality of the generated BIM model in terms of object-level completeness and geometric consistency.

4.2.1. Corresponding object matching between floor plans and elevations

The fourth step evaluated the accuracy of matching corresponding objects between floor plans and elevations. Using the centroid positions of detected openings, a cost-matrix formulation with the Hungarian algorithm was applied to establish one-to-one correspondences. This evaluation measured how accurately the method could align the same architectural elements across two-dimensional views without relying on CAD metadata or manual alignment. The corresponding object matching experiments used drawings from eight apartment buildings that contained both floor plans and elevations. A total of 3458 openings (3410 windows and 48 doors) were included in the analysis. Each object was manually labeled to create a ground-truth dataset for matching evaluation.

4.2.2. Automated BIM generation

The final step evaluated the performance of automated BIM generation in two aspects: (1) object-level evaluation and (2) volume-level evaluation. These evaluations were designed to verify whether the system could not only generate the required building components but also reproduce their geometric properties with spatial consistency. For this experiment, a BIM was created in Autodesk Revit using the outputs of all previous steps. The evaluation used drawings from buildings for which ground-truth BIM was manually created. To enable a consistent comparison, both the generated BIM and the ground-truth model were exported to IFC format. Architectural elements were extracted from each IFC file, and corresponding elements were matched based on their semantic category and placement. Object-level evaluation measured completeness by comparing the number of walls, doors, and windows in the generated BIM against the ground-truth BIM. Volume-level evaluation quantified geometric accuracy by comparing the solid volumes of corresponding BIM elements in the generated BIM and the ground-truth BIM, both per element category and overall.

4.3. Evaluation metrics

1) Object recognition evaluation

The performance of architectural object recognition in floor plans and elevations was evaluated using four metrics: Precision, Recall, F1-score, and mean Average Precision (mAP)@0.5. Precision measures the proportion of correctly detected objects among all detections, while Recall measures the proportion of correctly detected objects among all ground-truth instances. These are defined as follows:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (9)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (10)$$

Table 3

Hyperparameters of drawing object recognition model training.

Hyperparameter	Floor plan recognition	Elevation recognition
Epochs	100	100
Batch size	16	16
Optimizer	SGD	AdamW
Loss function	Focal loss	Focal loss
Image input size	1280*1280 pixels	1024*1024 pixels
Momentum	0.937	0.937
Learning rate	0.01	0.01
Weight decay	0.0005	0.0005

where True Positive (TP) indicates an architectural element correctly detected and classified into its true category (e.g., wall, door, window), False Positive (FP) denotes a non-existent or misclassified detection, and False Negative (FN) refers to an existing object that was missed by the model. To evaluate both classification and localization accuracy, the mAP@0.5 was computed for standard detection metric. Together, these metrics provide a comprehensive evaluation of the model's precision and spatial consistency under different dataset compositions and inference strategies.

2) Floor-wise clustering evaluation

The performance of floor-wise clustering in elevation drawings was assessed using three cluster validation indices: Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), and Homogeneity Score. The ARI quantifies the similarity between predicted clusters and ground-truth floor labels while adjusting for random chance, defined as:

$$ARI = \frac{RI - Expected(RI)}{max(RI) - Expected(RI)} \quad (11)$$

where RI denotes the Rand Index, representing the proportion of object pairs that are consistently clustered together or separately in both partitions. The NMI measures the shared information between predicted and true clusters:

$$NMI = \frac{2I(X, Y)}{H(X) + H(Y)} \quad (12)$$

where $I(X, Y)$ is the mutual information between the predicted cluster set X and the ground-truth Y , and $H(X)$ and $H(Y)$ are their respective entropies. Finally, the Homogeneity Score evaluates whether each predicted cluster contains only samples from a single ground-truth class:

$$Homogeneity\ Score = 1 - \frac{H(C|K)}{H(C)} \quad (13)$$

where $H(C|K)$ represents the conditional entropy of the classes given the clusters. These indices collectively assess the clustering's accuracy, consistency, and purity in distinguishing multi-story structures and floor boundaries from elevation objects.

3) Scale information recognition evaluation

The recognition of scale information was evaluated in two sequential phases: (1) detection of scale annotations and (2) recognition of numeric scale values. Both phases were evaluated using Accuracy, Precision, Recall, and F1-score, defined as:

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad (14)$$

$$F1\ score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (15)$$

Here, TP refers to a correctly detected or correctly interpreted scale annotation, FP to an incorrectly identified or misread annotation, FN to a missed annotation, and True Negative (TN) to the correct identification of an image without scale information. Accuracy provides an overall measure of correctness, while Precision and Recall capture the reliability and completeness of the recognition process. The F1-score summarizes both in a single metric. These measures were used to compare the performance of a LMM with traditional OCR engines such as EasyOCR and Tesseract, highlighting the LMM's superior contextual understanding and robustness in interpreting scales across diverse annotation styles.

4) Corresponding object matching evaluation

The accuracy of matching corresponding objects between floor plans and elevations was evaluated using Precision, Recall, and F1-score. The definitions of Precision and Recall remain consistent with drawing recognition, while the F1-score provides a harmonic meaning of both. In this experiment, TP corresponds to a correctly matched pair of floor plan and elevation objects representing the same architectural element, FP denotes an incorrect match, and FN indicates a missed correspondence. These metrics collectively measure the reliability of the Hungarian algorithm in establishing accurate one-to-one correspondences between two-dimensional representations without relying on CAD metadata or manual adjustments.

5) Automated BIM generation evaluation

The performance of automated BIM generation was evaluated using two metrics: (1) object-level metrics and (2) volume-level metrics. Object-level metrics measured whether the required architectural components were successfully generated by comparing the number of walls, doors, and windows in the generated BIM with those in the ground-truth BIM. Generation completeness for each element was defined as:

$$Generation\ Completeness = \frac{Number\ of\ generated\ elements}{Number\ of\ ground - truth\ elements} \quad (16)$$

Volume-level metrics measured geometric accuracy using Volumetric Intersection over Union (vIoU). vIoU quantifies the overlap between the solid volumes of corresponding elements in the generated BIM and ground-truth BIM, defined as:

$$Volumetric\ IoU = \frac{Volume(S_{gen} \cap S_{gt})}{Volume(S_{gen} \cup S_{gt})} \quad (17)$$

where S_{gen} and S_{gt} denote the solid geometry of the generated BIM and the ground-truth BIM, respectively. A higher IoU indicates better geometric fidelity and spatial consistency between the generated and ground-truth BIMs. This metric serves as a direct quantitative indicator of the accuracy and completeness of the 3D generation, reflecting how accurately the proposed system generates 3D building geometry from 2D architectural drawings.

5. Results

This section reports the experimental results of the proposed method and presents a comprehensive evaluation of its performance across all stages of the BIM generation pipeline.

Table 4

Comparison of recognition performance under different dataset and inference configurations.

Configuration	Precision	Recall	F1-score	mAP _{0.5}
FP_original (Baseline)	0.632	0.519	0.570	0.507
FP_original SAHI	0.645	0.608	0.626	0.556
FP_tiled	0.899	0.857	0.877	0.908
FP_tiled SAHI	0.905	0.889	0.897	0.934
FP_combined	0.625	0.524	0.570	0.512
FP_combined SAHI	0.638	0.601	0.619	0.563
ELEV_original (Baseline)	0.577	0.344	0.431	0.355
ELEV_original SAHI	0.565	0.472	0.514	0.398
ELEV_tiled	0.847	0.710	0.772	0.779
ELEV_tiled SAHI	0.835	0.784	0.808	0.812
ELEV_combined	0.906	0.764	0.829	0.819
ELEV_combined SAHI	0.888	0.806	0.845	0.838

FP = Floor plan, ELEV = Elevation, SAHI = Slicing Aided Hyper Inference. Baseline is defined as FP_original, ELEV_original.

5.1. Object recognition performance

Table 4 summarizes the performance achieved by each model configuration across different dataset compositions and inference strategies for both floor plans and elevations. The results demonstrate clear superiority of the proposed method over baseline approaches.

For floor plans, the proposed method (FP_tiled_SAHl) achieved mAP@0.5 of 0.934, representing a 41.8% improvement over the baseline model (FP_original, 0.507) trained on full-size images without inference optimization. The baseline model suffered from poor localization of small objects and struggled with dense spatial arrangements. When SAHI was applied to the baseline at inference time (FP_original_SAHl), performance improved to 0.556, but remained 67.9% lower than the proposed method, confirming that dataset design is more critical than inference-time augmentation alone. The proposed tiled training strategy combined with SAHI inference yielded Precision of 0.905, Recall of 0.889, and F1-score of 0.897, substantially outperforming the baseline across all metrics.

For elevations, the proposed method (ELEV_combined_SAHl) achieved mAP@0.5 of 0.838, **16.2% higher** than the baseline (ELEV_original, 0.358). The baseline model trained on full-size elevation images performed poorly due to disproportionate aspect ratios and insufficient coverage of architectural details. The combined dataset approach addressed these representation inconsistencies, improving detection and refining localization of small openings. As shown in Fig. 12, SAHI inference provided additional gains by enabling more precise instance segmentation compared to standard YOLO11s-seg processing. In areas with dense object clusters and overlapping elements, the proposed method reduced false positives and achieved more clearly separated object recognition, demonstrating its robustness in handling complex architectural drawings with missing or inconsistent dimensional information.

These performance improvements are visually demonstrated in Fig. 13. The baseline model (YOLO11-only) exhibited significant fragmentation of large facade elements such as walls, splitting them into multiple disconnected detections. This fragmentation directly contributed to the baseline's poor Precision scores reported in Table 4. In contrast, the proposed SAHI-enhanced method produced coherent, accurately bounded detections of wall elements while simultaneously improving the recognition of small objects such as windows and doors, which were frequently under-detected or missed entirely by the baseline

approach in elevations.

Fig. 13 illustrates the stark difference between baseline YOLO11-only recognition (c, g) and the proposed SAHI-enhanced recognition (d, h) for elevation drawings. In the upper elevation example, the baseline fragmented wall boundaries and failed to detect several window instances, whereas the SAHI-enhanced method achieved complete wall detection and identified all window openings. The lower elevation example further demonstrates this superiority: the baseline produced inconsistent door and window detections with irregular bounding boxes, while the SAHI-enhanced approach delivered uniform, accurate detections across the entire facade. These visual results corroborate the quantitative improvements shown in Table 4, where the proposed method achieved 0.838 mAP@0.5 compared to the baseline's 0.358—a direct consequence of eliminating fragmentation errors and improving small object detection.

Across both drawing types, the comparative results confirm that the proposed method's superiority stems from two synergistic factors: tailored dataset configuration aligned with drawing characteristics, and SAHI inference optimization. The substantial performance gains validate that the proposed method effectively addresses the fundamental limitations of conventional single-strategy approaches in automated architectural drawing recognition.

5.2. Floor-wise clustering performance

The proposed floor-wise clustering approach accurately recovered story-level structure from elevation drawings. When applied to 100 facades, the clustering results achieved an ARI of 0.9524, an NMI of 0.9656, and a Homogeneity score of 0.9652. These results indicate that the automatically derived clusters closely matched manually labeled floor boundaries, with minimal confusion between levels.

This is notable because multi-family residential facades do not always present perfectly regular vertical repetition. Variations in balcony shapes, window arrangements, or local facade modifications can obscure story boundaries. The high agreement between predicted clusters and ground-truth suggests that simple geometric cues, the vertical distribution of window and door centroids, are sufficient to infer the underlying building story structure without explicit metadata. This step is essential in our pipeline because it allows the system to relate floor plan information to the correct elevation context in a floor-aware manner.

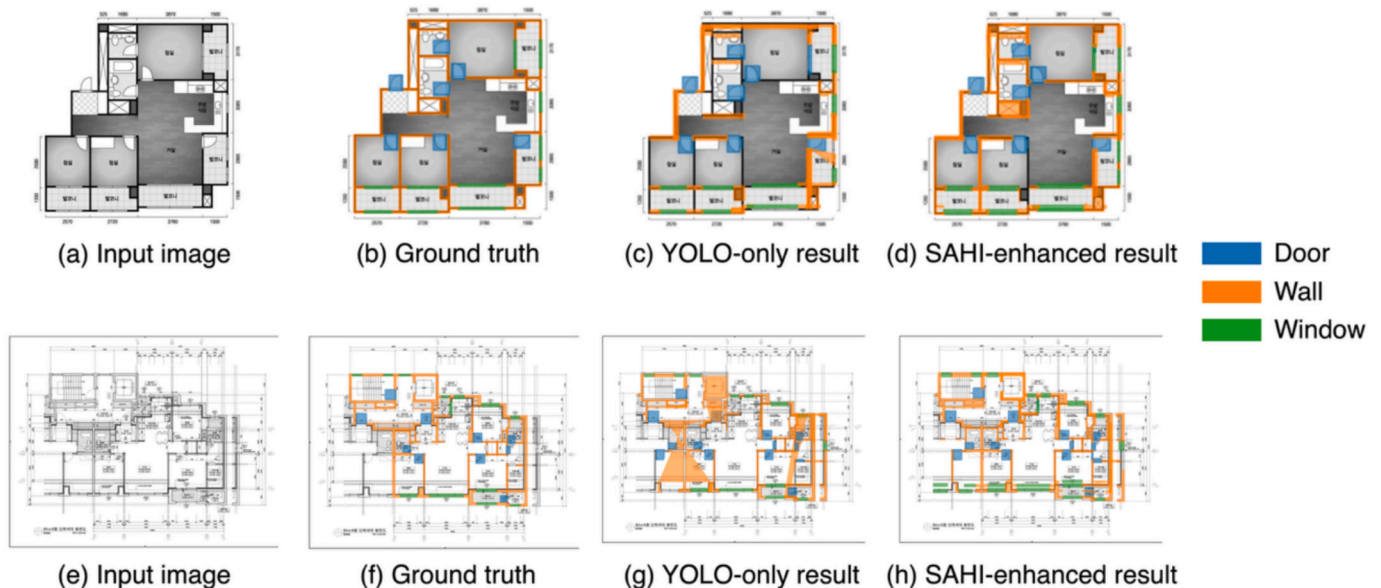


Fig. 12. Comparison between YOLO-only recognition and SAHI-enhanced recognition (Floor plan).

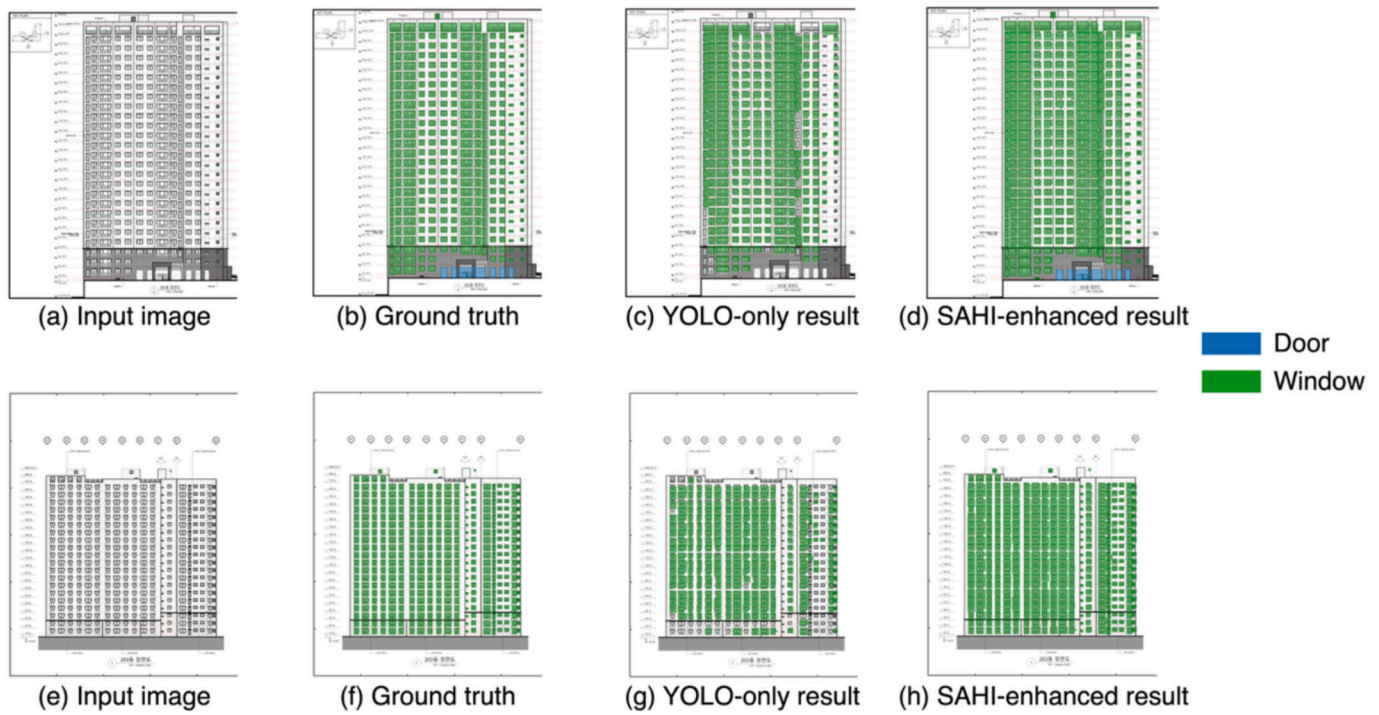


Fig. 13. Comparison between YOLO-only recognition and SAHI-enhanced recognition (Elevation).

5.3. Scale information extraction performance

Scale interpretation was tested in two stages: identifying whether a drawing contains scale information at all, and then extracting the correct numeric value. GPT-4o outperformed conventional OCR engines in both stages. In the first stage, GPT-4o achieved an Accuracy of 0.962, Recall of 0.969, and F1-score of 0.980 (Fig. 14). EasyOCR and Tesseract exhibited high Precision but substantially lower Recall, meaning they often failed to detect scale information even when it was present.

In the second stage, GPT-4o again achieved the highest performance with an Accuracy of 0.707, compared to 0.198 for EasyOCR and 0.369 for Tesseract (Fig. 15). This indicates that, beyond simply finding that a scale exists, GPT-4o was also more successful at extracting the correct scale information.

These results show that multimodal reasoning is effective in

interpreting architectural annotations that are noisy, incomplete, or represented in non-standard formats. This capability is critical for automated BIM generation, because converting pixel coordinates into metric dimensions depends on reliable access to scale information. In many legacy drawings, this information is either inconsistently annotated or partially missing.

In conclusion, this experiment confirms that multimodal reasoning with LMMs offers a more effective approach for robust scale information recognition in architectural drawings, compared to conventional OCR-based methods.

5.4. Corresponding object matching performance

We evaluated whether the system could correctly align architectural elements across floor plans and elevations, a requirement for creating 3D

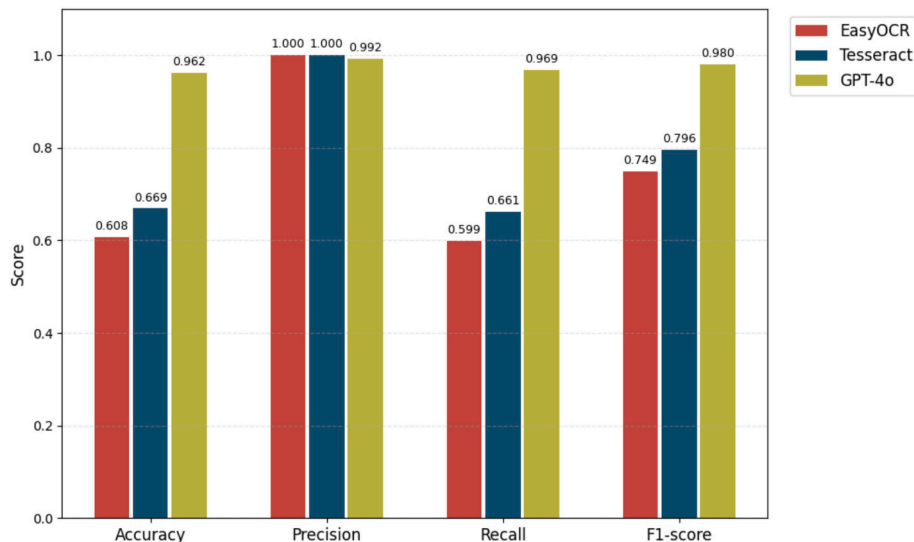


Fig. 14. Experiment result of stage 1: Comparison of scale presence detection.

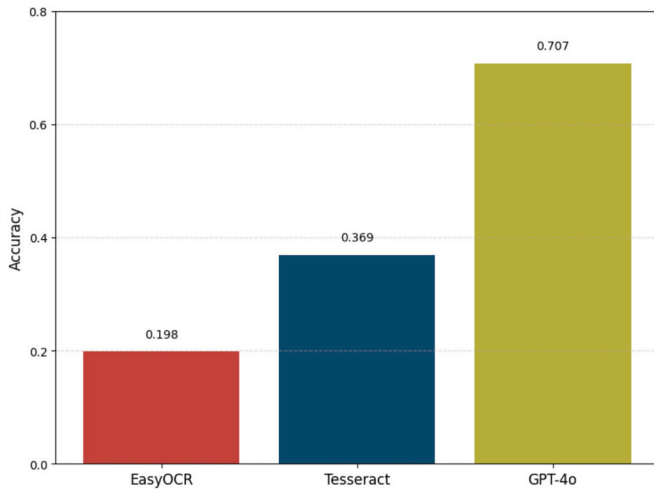


Fig. 15. Experiment result of stage 2: Comparison of scale value recognition accuracy.

geometry from 2D sources. Using drawings from eight buildings (3458 total objects: 3410 windows and 48 doors), the proposed matching algorithm achieved a Precision of 0.913, Recall of 0.949, and F1-score of 0.93 for windows. For doors, performance was even higher, with a Precision of 0.965, Recall of 0.976, and F1-score of 0.95 (Table 5).

The higher scores for doors can be explained by their relatively small number and clearer geometric distinctiveness, which reduces ambiguity during matching. Most errors arose from cases where detections in the two drawings were inconsistent. For example, a window detected in the elevation but fragmented or missing in the plan. In such cases, multiple plan objects were sometimes associated with a single elevation object, which lowered the matching Precision. Even with these challenges, the overall performance shows that reliable cross-view correspondence can be established without relying on CAD metadata or manual alignment, which is a core requirement for automated BIM generation from raster drawings.

5.5. BIM generation performance

The performance of automated BIM generation was evaluated using object-level generation completeness and volume-based geometric accuracy, which together reflect the quality of the final BIM output in terms of component completeness and geometric fidelity.

First, object-level results were examined by comparing the number of generated elements with those in the ground-truth BIM. As shown in Table 6, the generated BIM contained 135 walls, 64 doors, and 180 windows, resulting in completeness values of 0.882, 1.000, and 0.952, respectively, and an overall completeness of 0.934. These results indicate that the proposed method generated most required architectural components with high completeness.

Next, volume-level geometric accuracy was evaluated using vIoU. Table 7 presents the volumetric comparison for each element category. Wall and door elements achieved vIoU values of 0.695 and 0.626, respectively, indicating substantial geometric consistency in terms of volume and placement. Window elements obtained a lower vIoU value of 0.364, which can be attributed to their relatively small volumes and higher sensitivity to minor positional deviations. Despite this, the

Table 5
Corresponding object matching performance.

Building object	Precision	Recall	F1-score
Window	0.913	0.948	0.93
Door	0.965	0.976	0.95

Table 6
Object-level generation completeness of the generated BIM.

Element	Ground-truth count	Generated count	Generation Completeness
Wall	153	135	0.882
Door	64	64	1.000
Window	189	180	0.952
Total	406	379	0.934

Table 7
Volumetric IoU between the generated BIM and the ground-truth BIM.

Element	Ground-truth volume (m ³)	Generated volume (m ³)	Union volume (m ³)	Intersection volume (m ³)	Volumetric IoU
Wall	2441.45	2460.86	2892.69	2009.62	0.695
Door	7.67	8.00	9.64	6.03	0.626
Window	11.69	10.40	16.20	5.89	0.364
Overall	2460.81	2479.26	2917.76	2022.31	0.693

overall vIoU of 0.693 suggests that the generated BIM preserves the global building geometry and spatial configuration with a high degree of volumetric consistency. Overall, the object-level and volume-level results demonstrate that the proposed method generates a coherent and geometrically consistent BIM from 2D drawings.

Fig. 16 presents a qualitative comparison between the generated BIM model and the ground-truth BIM. The generated BIM successfully reconstructs the overall building configuration and major architectural components, including walls, doors, and windows, across multiple floors. The spatial arrangement and relative placement of elements are visually consistent with the ground-truth model. While lower vIoU values were observed for window and door elements, the qualitative comparison suggests that their object counts and horizontal placement do not exhibit pronounced deviations from the ground-truth BIM. The enlarged views (Fig. 16 (c), (d)) further highlight local geometric differences while confirming that the global structure and component relationships are preserved, providing visual support for the quantitative evaluation results.

6. Discussion

Based on an analysis of the experimental results, this section interprets the findings and discusses key factors that affect the performance and applicability of the proposed method. The discussion is organized around two main themes: (1) key findings from experimental results and (2) implications and limitations.

6.1. Influence of drawing characteristics on optimal dataset and inference strategies

The proposed method is designed to operate within a defined scope that aligns with the structural and geometric characteristics of apartment-type architectural drawings. Within this scope, variations in drawing complexity primarily arise from differences in spatial layout, object density, and the relative importance of global context versus local detail. The following discussion analyzes how these variations influence recognition performance and motivate different dataset configurations and inference strategies.

The evaluation results revealed that the optimal combination of dataset configuration and inference strategy depends strongly on the visual and geometric characteristics of the drawing. Specifically, the model trained on the tiled dataset achieved the highest performance for floor plan recognition, while the combined dataset yielded the best results for elevation recognition. This result highlights that data preparation and inference strategies are not merely implementation choices

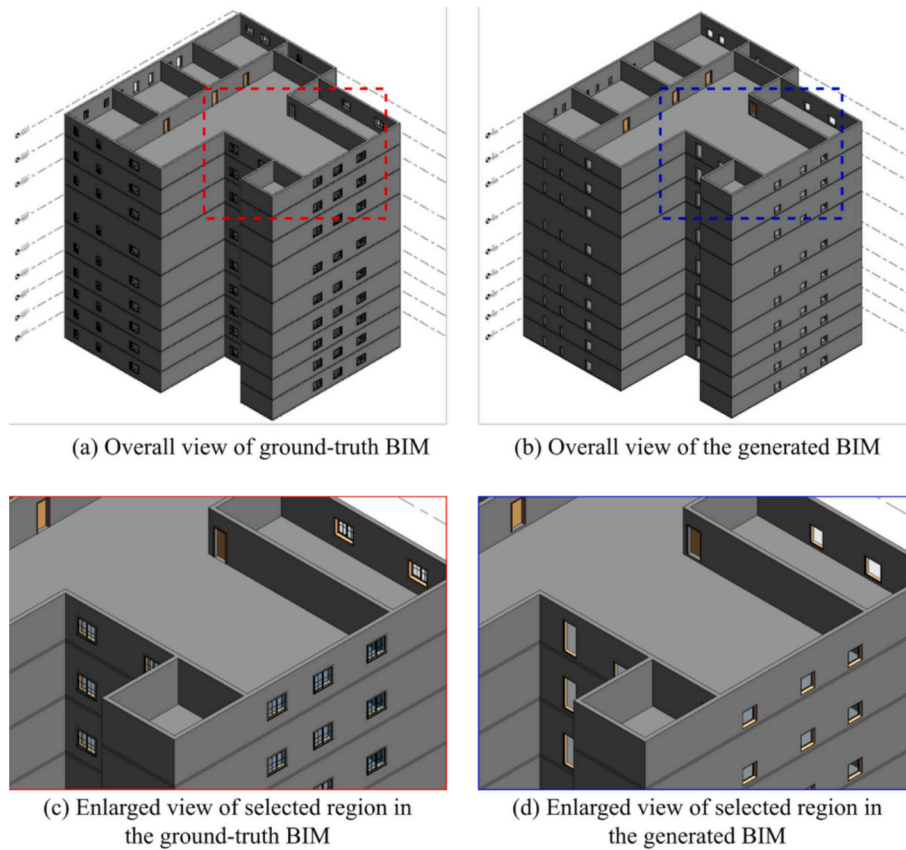


Fig. 16. Qualitative comparison between the generated BIM and the ground-truth BIM.

but play a critical role in handling layout complexity inherent in different drawing representations.

From a visual and geometric perspective, floor plans exhibit predominantly planar characteristics, where architectural elements such as doors and windows are distributed within a single plane. In floor plans with increased layout complexity—characterized by dense object distributions, small-scale components, and fragmented local geometry—recognition performance is often compromised by the relative loss of resolution for small features, leading to higher false-negative rates in standard detection pipelines. In such cases, recognition performance is largely governed by the model's ability to capture fine-grained visual details rather than global spatial context. Subdividing drawings into smaller patches enhances local feature extraction by enlarging the relative scale of small objects within each tile, enabling more robust detection in visually dense layouts. In addition, applying SAHI inference further mitigates complexity-related challenges by ensuring that small or partially occluded objects near tile boundaries are effectively detected. The combination of tiled training and SAHI inference therefore provides complementary advantages. Tiling strengthens feature learning at the training stage, while SAHI reinforces small object visibility during inference. Together, they form the most effective configuration for drawings dominated by small and densely distributed components.

In contrast, elevation drawings present a different form of layout complexity, where both global building context and local details are simultaneously important. Elevations encode vertically extended geometry with repeated facade patterns across floors, and excessive fragmentation through tiled-only training can disrupt these global-to-local dependencies. In highly complex layouts where architectural elements are densely packed or non-standardized, the risk of missing global vertical patterns or misidentifying repeated elements increases. While full-size training preserves vertical continuity and repetitive structure, it

may lack sufficient resolution for accurately detecting small façade elements. The combined dataset addresses this trade-off by integrating full-size images for contextual learning with tiled images for local detail enhancement. When paired with SAHI inference, this approach improves robustness in complex facade layouts by maintaining vertical coherence while still capturing small-scale openings. Consequently, the combined training and inference strategy is better suited for elevation drawings in which layout complexity arises from repetitive vertical structures and strong global-to-local relationships.

Overall, these findings suggest that training and inference strategies should be selected based on the dominant visual characteristics of the drawing rather than its nominal type. Drawings with planar characteristics, featuring small and densely distributed components with limited global context, benefit most from tiled training with SAHI inference. Drawings with elevational characteristics, featuring vertically extended geometry and strong global-to-local dependencies, are better handled with combined training and SAHI inference. This classification framework can also be applied to drawings with ambiguous or mixed characteristics, such as elevations of single-story buildings that exhibit visual characteristics similar to floor plans or floor plans that contain long and continuous linear elements resembling elevations. By interpreting drawings according to their spatial and geometric characteristics rather than their predefined categories, the proposed strategy provides a generalized guideline for dataset preparation and inference configuration in architectural drawing recognition tasks.

6.2. Potential of large multimodal models for dimensional information recognition

The results of the dimensional information recognition experiment highlight the emerging potential of LMMs to interpret dimensional information in architectural drawings. Traditional OCR-based methods

are limited by their reliance on explicit textual features and predefined font patterns. In contrast, LMMs can reason across both visual and textual domains, enabling them to understand the contextual meaning of scale annotations even when they are incomplete, stylized, or inconsistently represented. This suggests that the recognition process is no longer constrained by the formal structure of text but can instead rely on semantic and spatial reasoning that mirrors human interpretation.

This capability has particular significance for architectural drawings, where scale and height annotations are often irregular, overlapping, or visually integrated with other graphical elements. By simultaneously processing the spatial layout and linguistic context, LMMs can infer measurement intent rather than merely extracting characters. For instance, when a scale bar is partially occluded or when numeric ratios are replaced with descriptive labels, the model can still derive the intended metric relationship by reasoning about nearby visual and textual cues. This contextual understanding enables robust interpretation of ambiguous annotations that conventional OCR cannot reliably handle.

In a broader sense, these findings suggest that LMMs can bridge the gap between geometric recognition and semantic understanding in automated BIM generation. Their ability to interpret both visual and textual cues within complex drawings enables the integration of scale, height, and other contextual information without rule-based pre-processing. As these models continue to advance, they hold the potential to enable BIM generation systems that operate with minimal human intervention, transforming architectural drawings from static geometric depictions into semantically rich sources of structured building information.

Nevertheless, the effectiveness of LMM-based dimensional information recognition is influenced by the availability and clarity of contextual cues within the drawing. In cases where annotations are entirely absent, highly unconventional, or locally defined without surrounding semantic context, reliable inference remains challenging. Therefore, while LMMs significantly enhance robustness compared to conventional OCR-based approaches, their performance remains dependent on the presence of interpretable visual cues and should be interpreted within the scope of typical architectural drafting practices.

6.3. Trade-off between information reliability and geometric consistency in automated BIM generation

The BIM generation experiment revealed a clear distinction between global geometric consistency and local spatial fidelity. At the object level, the proposed system successfully generated most architectural components, achieving over 93% completeness, indicating that the integration of floor plan and elevation information provides sufficient semantic consistency to determine the existence and general placement of key elements in three-dimensional space.

Volume-level evaluation results further confirmed that the generated BIM preserves global building structure. Wall elements achieved a vIoU of 0.695, demonstrating a high level of agreement in overall massing, wall layout, and story stacking. Door elements also showed a relatively high vIoU value of 0.626. However, this does not necessarily indicate high geometric precision, as doors occupy relatively small volumes and their placement is primarily governed by the alignment of vertical coordinates between plans and elevations. When floor-to-floor heights and sill positions are correctly inferred, door geometry can align reliably even with minor horizontal deviations. In addition, the vIoU of wall elements may also be affected by vertical misalignments propagated from hosted window and door elements during BIM instantiation, rather than indicating systematic errors in wall layout reconstruction itself.

It should be noted that vIoU is inherently sensitive to small geometric deviations for objects with limited volume, such as windows. In such cases, even minor errors in sill height or head height can result in a substantial reduction in volumetric overlap. In the proposed method, vertical coordinates of window and door elements extracted from

elevation drawings are unified at the cluster level after instance segmentation and floor-wise clustering to ensure consistency across floors. While this strategy improves robustness at the story level, it may smooth out instance-level variations and introduce small deviations in individual element heights. From this perspective, a vIoU value of 0.364 for window elements represents a relatively strong geometric agreement given the geometric sensitivity of these components. This observation also highlights a limitation of vIoU as an evaluation metric for fine-grained building elements. While vIoU is effective for assessing global geometric consistency at the building or story level, it may underestimate reconstruction quality for small-scale architectural components. In addition, BIM element instantiation involves geometric simplification and coordinate adjustment, which can further reduce volumetric overlap for small elements. These discrepancies stem from the informational limitations of unstructured 2D drawings, including incomplete height annotations, inconsistent projection lines, and minor distortions in elevation representations.

Within this context, it is important to clarify the intended quality scope of the generated BIM. The objective of this study is not to reproduce fabrication-level geometric precision, but to generate a building-scale BIM that preserves semantic completeness and volumetric consistency across floors. From this perspective, the obtained quantitative results, together with the qualitative comparison presented in Fig. 16, demonstrate that the proposed method produces a BIM of sufficient quality for applications such as building massing analysis, energy simulation, and high-level spatial assessment. Ultimately, this underscores a fundamental trade-off in automated BIM generation from unstructured 2D drawings: while global geometric consistency can be achieved through multi-view drawing integration, local accuracy remains critically dependent on the reliability and precision of the underlying source information.

6.4. Limitations and future work

This study has several limitations. First, the scope of object recognition in this study is limited to three architectural elements: walls, windows, and doors. While real-world buildings often include additional components such as columns, beams, and staircases, the selected object categories are highly representative in the context of apartment-type buildings. In such buildings, most architectural layouts can be effectively captured using walls, windows, and doors alone. Therefore, the current object categories are considered sufficient for evaluating the core feasibility and accuracy of the proposed method. Nevertheless, the generalizability of the model may be limited when applied to buildings with more diverse structural components or unconventional drawing representations. Second, the proposed method is based on the Manhattan-world assumption [32], which implies that all objects are orthogonally aligned and that the drawing is not rotated. This assumption is applied during the post-processing stage to simplify and align geometry. However, in real-world scenarios, drawings may depict buildings with rotated or non-rectilinear geometries, making the current approach difficult to apply in such cases. Third, regarding the recognition of dimensional information, the performance of LMMs is critically dependent on the presence and clarity of contextual cues within the drawing. As discussed in Section 6.2, in cases where scale and height annotations are entirely absent or highly unconventional, reliable inference remains challenging. This suggests that while LMMs enhance robustness compared to conventional OCR, they are not yet fully autonomous and function best within the scope of standard architectural drafting practices. Fourth, the study relies on the integration of floor plans and elevation drawings to generate 3D geometry. While this is effective for estimating the height of exterior elements, interior components such as internal doors are not visible in elevations. Consequently, the local spatial fidelity of interior elements remains limited, as direct vertical information is unavailable in elevation drawings. Taken together, these limitations indicate that the quality of the generated BIM

is inherently bounded by the completeness and reliability of the input drawings and the underlying modeling assumptions, rather than by the BIM generation process itself.

To address these limitations, several future research directions are proposed. First, to increase the generalizability and completeness of BIM generation, future work should consider expanding the object categories to include additional architectural elements such as columns, beams, and staircases. In parallel, the training dataset should be diversified to include different types of buildings and variations in drawing conventions, thereby enhancing model robustness and enabling broader applicability in practice. Second, we should develop post-processing algorithms capable of handling rotated or non-orthogonal drawings. By integrating rotation correction techniques or supporting object detection along arbitrary orientations, the approach can be made more flexible and applicable to irregular or complex designs. Finally, future work should incorporate section drawings. This would allow for the generation of more complete BIMs that include both exterior and interior architectural elements, improving the comprehensiveness and usability of the output.

7. Conclusions

This paper presented an automated pipeline for generating BIMs directly from floor plans and elevations. The focus of this research was to address the informational gaps in 2D legacy drawings, specifically within the standardized architectural layouts of apartment-type buildings. The proposed method consists of two main phases: (1) extracting geometric and semantic information from 2D drawings and (2) generating 3D BIM through corresponding object matching and BIM element instantiation. Experimental results indicate that the proposed method achieves strong performance across drawing object recognition, scale recognition, and cross-view matching, enabling consistent building-scale BIM generation from floor plans and elevations.

Previous studies on automated BIM generation have revealed several research gaps, such as the inability to extract vertical information resulting from the use of a single drawing type, the lack of dataset strategies considering the visual and geometric characteristics of each drawing type, the dependency on unreliable OCR-based dimensional interpretation, and the absence of explicit object-level correspondence between floor plans and elevations. This study addressed these limitations by introducing a unified and data-driven pipeline that integrates complementary geometric information from both floor plans and elevations.

The main contributions of this study are as follows. First, we identified that the optimal configuration of dataset composition and inference strategy is inherently dependent on the visual and geometric characteristics of each drawing type. By adopting tiled datasets with SAHI for floor plans and combined datasets with SAHI for elevations, the proposed method achieved mAP@0.5 scores of 0.934 and 0.838, representing 41.8% and 16.2% improvements over baseline configurations. These results suggest that dataset composition and inference strategy selection play an important methodological role, rather than serving merely as preprocessing choices. Second, an LMM-based reasoning framework was introduced to extract and interpret scale and dimensional annotations even under noisy or incomplete conditions. The approach achieved 0.707 accuracy in scale recognition, outperforming traditional OCR engines. This suggests that semantic reasoning can effectively complement visual perception, offering a more robust alternative to traditional OCR engines for metric calibration in unstructured drawings. Third, this study introduced an explicit cross-view correspondence mechanism that integrates complementary geometric cues between the two drawing types. The matching process, formulated as a cost-matrix optimization and solved using the Hungarian algorithm, achieved a high correspondence F1-score of 0.93. This approach addresses the often-overlooked challenge of geometric consistency between heterogeneous 2D representations and provides a systematic basis

for integrated BIM generation.

From a practical standpoint, the proposed method contributes to the digital transformation of the construction industry by enabling preliminary BIM generation from 2D legacy drawings. While it may not yet replace manual detailed modeling for complex structures, it can effectively support building lifecycle management, retrofit planning, and archival documentation. Moreover, the system is designed to operate robustly under imperfect drawing conditions, such as overlapping symbols, missing dimensions, and varied annotation conventions, demonstrating adaptability to real-world architectural datasets.

Despite these contributions, certain limitations remain. The current recognition process focuses on three object types: walls, windows, and doors, which are sufficient for apartment-type buildings but restrict applicability to more complex structures. In addition, the geometric generation process assumes orthogonal alignment, which limits robustness when applied to drawings containing rotated or non-rectilinear geometries. Furthermore, because the proposed method relies on the integration of floor plans and elevations, interior components that are not visible in elevations cannot be fully generated in three dimensions. As a result, the quality of the generated BIM remains inherently dependent on the completeness and reliability of the input drawings, and the method performs most effectively within the scope of typical architectural drafting conventions. We will address these limitations in future work by expanding the set of recognized object categories and diversifying the training dataset to include a broader range of building types and drawing styles, thereby improving generalizability. In addition, developing post-processing techniques that support arbitrary orientations could improve robustness. Finally, incorporating section drawings would enable height estimation for interior components and support more complete generation of both exterior and interior architectural features.

Overall, this study demonstrates that automated BIM generation from architectural drawings is both technically feasible and practically scalable. By combining data-driven drawing recognition with multi-modal reasoning, the proposed method provides a methodological foundation for information-consistent BIM generation from heterogeneous 2D drawings, enabling coherent integration of geometric and semantic information across multi-view drawings.

Declaration of generative AI use

During the preparation of this work the authors used an LLM in order to proofread the manuscript. After using an LLM, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRedit authorship contribution statement

Hyunjun Lee: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Conceptualization. **Suhyung Jang:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation. **Junghun Lee:** Writing – review & editing, Methodology, Investigation. **H. David Jeong:** Writing – review & editing, Supervision, Funding acquisition. **Ghang Lee:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

This research used datasets from 'The Open AI Dataset Project (AI-Hub, S. Korea)', which were not created by the authors. All data information can be accessed through 'AI-Hub (www.aihub.or.kr)'.

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